INSY7315 WORK INTEGRATED LEARNING

# REEF Operations Platform User Manual

One connected system for maintenance, stock, production, purchasing and reporting, kept separate plant by plant. Version 2, which adds the handset application.

| | |
|---|---|
| SYSTEM | REEF Operations Platform |
| DOCUMENT | User Manual, version 2 |
| FOR | Plant workers, site managers and the owner |
| CLIENT | Resource Energy Engineering Fuels (REEF) |
| MODULE | INSY7315 Information Systems 3E |
| PREPARED BY | Bradley Rwafa, Group 3, Codal Reef |
| DATE | 1 November 2026 |

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# 1. Introduction

The REEF Operations Platform records what happens on a plant as it happens, and turns it into stock levels, costs and performance figures without anybody rewriting them. A worker records a repair, a fuel slip, a production figure, a stoppage or a part taken off the shelf on a screen instead of a paper slip. A site manager reads it back, keeps the registers correct, orders what the plant needs and watches what it costs. The owner sees every plant at once.

## 1.1 What the system is for

Before this system REEF ran on paper slips, personal spreadsheets and memory. There was no shared record of equipment, no live view of stock, and no way to answer what a plant cost per ton until long after the month had ended. The platform exists to close that gap. Its single purpose is that a figure is captured once, by the person who saw it, and is never typed a second time.

Everything else follows from that. The cost per ton on the dashboard is built from the tonnage a worker entered on a phone, the fuel slip they photographed, the parts a repair used and the salaries a manager keyed in once a month. Nobody assembles it.

## 1.2 Who this manual is for

| IF YOU ARE A | YOU USE THE SYSTEM TO | READ |
|---|---|---|
| **Plant worker** | Record production, downtime, repairs, fuel and stock taken off the shelf, on a phone or tablet at the plant, and see what is in the store | Sections 2 to 7, then 15 |
| **Site manager** | Read back what was captured, correct it, keep the registers, manage your plant's stock, approve and receive orders, and watch the cost per ton | All of it, and section 9 first |
| **Owner** | Everything a site manager does, at every plant, plus comparing plants, the records that carry personal information, who may sign in, and whether the system itself is running | All of it. The administrator guide covers the owner only screens |

## 1.3 How to read this manual

Every screen in this manual has three things: what it is for, the steps to use it, and a picture of it. The pictures are photographs of the running system with demonstration data in it, so what is printed here is what is on the screen rather than a drawing of it. The names, sites and figures in them are invented.

Three kinds of note appear throughout.

> **TIP**
> Something that makes the system quicker or easier to live with. Ignoring a tip costs you a little time and nothing else.

> **NOTE**
> Something about how the system behaves that is not obvious from the screen, or a rule REEF set. Worth knowing before you are surprised by it.

**WARNING**
Something that can cost REEF money, lose information, or break a rule the law sets. These are the ones to read twice.

## 1.4 What is new in version 2

Two things are new. Workers can now claim back money they spent out of their own pocket, which is section 6. And the system can be installed as an application on an Android phone, which is section 7. Version 1 of this manual was written when the system could only be reached in a web browser, and it said in three places that there was no application to install; those have been corrected, section 2 now says what the application needs, and the role table, the troubleshooting list, the questions and the glossary have all been brought into line. Nothing else about how the system works has changed, so if you have read version 1, sections 6 and 7 are the two to read.

# 2. What you need

There are two ways to reach the system. Both are the same system, with the same sign in, the same screens and the same rules: a web browser on any device, and an Android application a worker installs on a handset. Nobody has to choose. Most managers use a browser on a desktop and most workers use the application on a phone, because that is what suits each of them.

| REQUIREMENT | WHAT IS NEEDED | WHY |
|---|---|---|
| **Browser** | Chrome, Edge, Firefox or Safari, updated within the last two years | The screens use browser features older versions do not have. A very old browser may show a blank page rather than an error |
| **Screen** | Any size. The screens rearrange for a phone, a tablet or a desktop | Workers use a phone at the plant. Managers use a desktop in the office. It is one application, not two |
| **The handset application** | Android 7 or later, with about 10 MB free | Installed from the package your site manager gives you. There is no iPhone version in this release; an iPhone reaches the system in Safari instead |
| **Internet** | Required throughout, in the browser and in the application | Every screen reads and writes as you use it. See the warning below |
| **Device permissions** | The camera only, and only when you choose to attach a photo | The system never asks for your contacts, your microphone or your location |
| **Account** | An email address and password, with a role and a plant set for you by the owner | The role decides what you may do and the plant decides whose records you see. Section 4 explains both |

**WARNING**

Capture needs signal, in the application as well as in the browser. A part of the plant with no coverage is a part of the plant where you cannot capture. What the application does is tell you so, on a plain page with one button, instead of spinning and losing what you typed. Write the figures down and enter them when you are back in signal, the same day. Holding a whole shift on the handset until signal returns is the next thing being built, and section 7.5 says what it will do.

# 3. Getting started

## 3.1 Reaching the system

1. Open the browser on your phone, tablet or computer, or open the REEF application if it has been installed on your handset. Section 7 covers installing it.
2. Type the address your site manager gave you. It is the same address for everybody, and the application already knows it.
3. In a browser, add it to your home screen or bookmarks the first time, so you never type it again. On a phone, use the browser menu and choose Add to Home Screen.

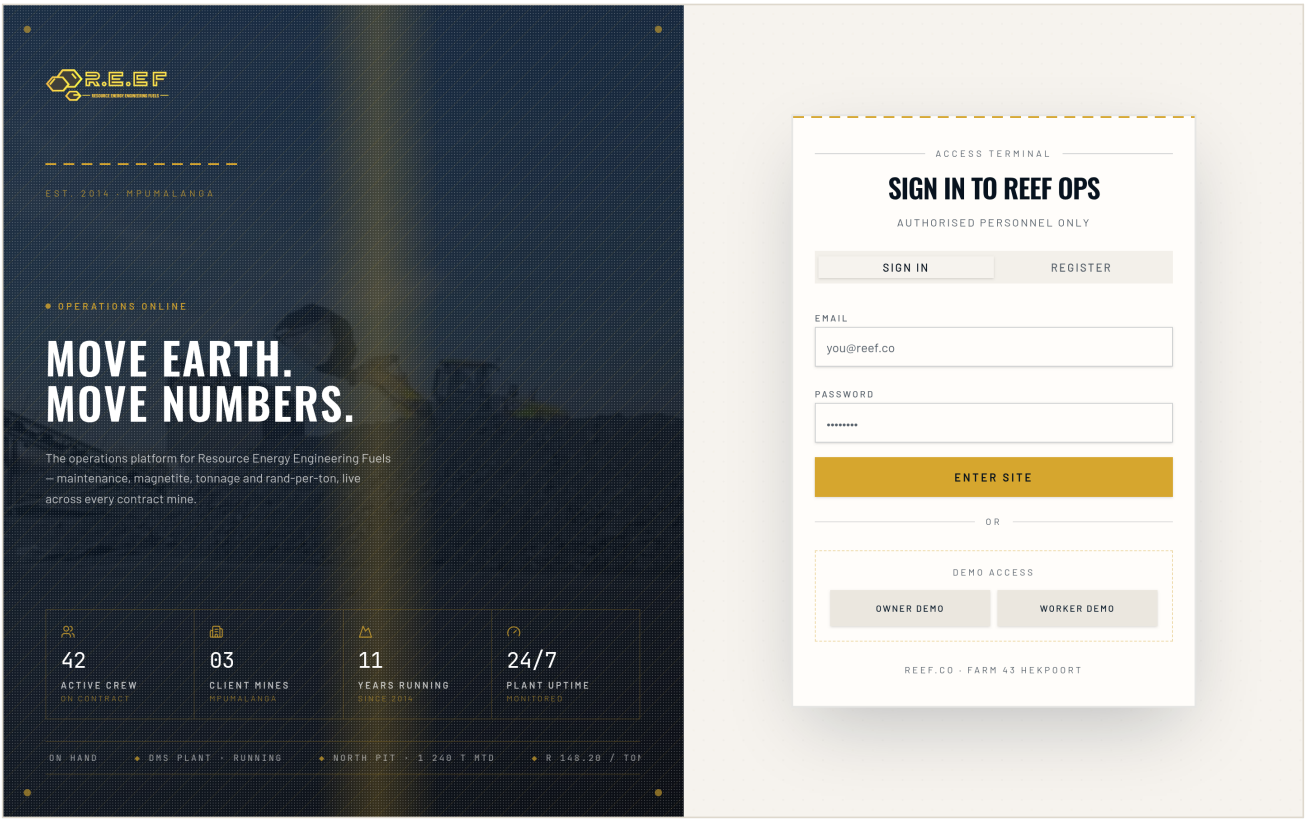



*Figure 1: The sign in screen. The form is on the right; the panel on the left is the same on every device.*

## 3.2 Signing in

1. Enter the email address registered for you, in Email.
2. Enter your password, in Password.
3. Select Enter site.
4. The system takes you to the screen that matches your role. A worker lands on the capture buttons and a manager lands on the dashboard, so the two never see each other's first screen.

NOTE

There is a Register tab, and it is not the way in. An account carries a role and a plant, and together those decide which money figures you see and whose stock you see. Ask your site manager to have an account created and set up for you, rather than registering yourself and ending up with no plant and the wrong role. A new account that registers itself can see almost nothing until the owner sets both.

## 3.3 Your plant

Every person except the owner belongs to one plant. You cannot change your own, and nor can your site manager. The database refuses it. That is not a lack of trust: your plant is what decides whose stock and whose purchase orders you can see, so letting people set their own would make the whole separation meaningless.

If your plant is wrong, your stock list looks empty or shows somebody else's parts. Tell the owner rather than working around it.

## 3.4 If you cannot get in

| WHAT YOU SEE | WHAT TO DO |
|---|---|
| **Your email or password is not recognised** | Check the email address for a typing mistake first, since it is the more common of the two. If it is right, ask the owner to reset your password. Nobody can read your existing password, so it is replaced rather than looked up |
| **Too many attempts, try again shortly** | Wait fifteen minutes. Repeated failed attempts are slowed on purpose, so somebody guessing at passwords cannot keep guessing quickly |
| **This sign in has been revoked. Sign in again** | Somebody, usually the owner, signed that device out. If it was not you who asked for it, say so: it is what happens when a phone is reported lost |
| **The page will not load at all** | Check that you have signal. If other websites load and this one does not, tell your site manager: it is the system and not your phone |

## 3.5 Signing out

On a desktop, Sign out is at the top right of every screen. On the worker screens it is the icon at the top right, and it is in the same place in the handset application. Signing out matters more on a shared plant phone than on your own: whatever is captured next is recorded against whoever is signed in, not whoever is holding it.

# 4. Roles, plants and what each allows

There are three roles, and separately everybody except the owner belongs to one plant. The role decides what kind of thing you may do. The plant decides whose records you may do it to. Both are set by the owner.

## 4.1 The three roles

| ROLE | WHO HOLDS IT | WHAT THEY ARE RESPONSIBLE FOR |
|---|---|---|
| **Worker** | Plant staff working on the equipment and the wash plant | Recording what happened while it is still fresh: the tons, the stoppage, the repair, the fuel and the parts taken off the shelf. Accuracy at the point of capture is the one thing nobody else can do for them |
| **Site manager** | The manager of one plant | Everything about their own plant: the registers, the equipment, the stock, the ordering, correcting a record that was captured wrongly, and watching the cost per ton |
| **Owner** | REEF's owner | Everything a site manager sees, at every plant, plus the things that cross plants: comparing them, who may sign in and as what, the records that carry personal information, the platform settings, and whether the system is still running |

## 4.2 What the system allows

This is the table the system enforces. Where a row is marked No, the screen does not offer it and the server refuses it even if the request is made another way, so hiding a button is a convenience and the refusal underneath it is the rule.

| WHAT THE SYSTEM ALLOWS | WORKER | MANAGER | OWNER |
|---|---|---|---|
| **Sign in, and see your own role and plant** | Yes | Yes | Yes |
| **See the list of sites** | Yes | Yes | Yes |
| **Add or change a site** | No | Yes | Yes |
| **Capture production, downtime, a repair or a fuel slip** | Yes | Yes | Yes |
| **Correct or delete a captured entry** | No | Yes | Yes |
| **Delete a fuel slip** | No | No | Yes |
| **See stock levels at your own plant** | Yes | Yes | Yes |
| **Book stock out of the store** | Yes | Yes | Yes |
| **Add or change a stock item, or its reorder levels** | No | Yes | Yes |
| **See another plant's stock, orders or reorder requests** | No | No | Yes |
| **Turn a reorder request into a draft order** | No | Yes | Yes |
| **Approve, send, receive or cancel a purchase order** | No | Yes | Yes |
| **Attach a photo to a capture** | Yes | Yes | Yes |
| **File a reimbursement claim for money spent out of pocket** | Yes | Yes | Yes |
| **See your own reimbursement claims** | Yes | Yes | Yes |
| **See other people's claims at your own plant** | No | Yes | Yes |
| **See every claim, at every plant** | No | No | Yes |
| **Mark a claim paid, or void it with a reason** | No | No | Yes |
| **Read the change history** | No | Yes | Yes |

| WHAT THE SYSTEM ALLOWS | WORKER | MANAGER | OWNER |
|---|---|---|---|
| **See cost per ton for a site** | No | Yes | Yes |
| **Compare sites against each other** | No | No | Yes |
| **Read the month end report and download the spreadsheet** | No | Yes | Yes |
| **Ask Reefie a question** | No | Yes | Yes |
| **See your own notifications and your own sign ins** | Yes | Yes | Yes |
| **See or sign out somebody else's devices** | No | No | Yes |
| **See an employee's identity number** | No | No | Yes |
| **See what is due for removal under the retention rules** | No | No | Yes |
| **See whether the scheduled reminders have been running** | No | No | Yes |
| **Change somebody's role** | No | No | Yes |
| **Read the platform settings** | Yes | Yes | Yes |
| **Change a platform setting** | No | No | Yes |

## 4.3 Six rules in that table that surprise people

| THE RULE | WHY IT IS THAT WAY |
|---|---|
| **Each plant sees only its own stock, its own reorder requests and its own orders** | REEF asked for this. Each plant has its own management team, and one team's figures are not the other team's business. The rule lives in the database, not on the screen, so it holds whichever way the record is reached |
| **A worker may book stock out but may not order more** | REEF asked for this too. Anybody may take a part off the shelf and record it, and anybody may see what is left. Booking a part out raises a reorder request when the figure runs low. Turning that request into an order spends REEF's money, so it stays with management |
| **Anybody may capture an entry, only a manager may correct one** | The person who saw it is the right person to record it. Changing a figure after the fact moves the cost per ton of a month that may already have been reported, which is a different kind of act and is recorded as one |
| **A site manager may not compare sites** | A manager sees every figure for their own site, including its cost per ton. The ranking across sites is the owner's, for the same reason REEF gave about plants |
| **A site manager may not see identity numbers** | An identity number is needed only for payroll and for the records the law requires. The fewer people who can read one, the fewer there are to account for if one is misused. Every look at one is recorded, including a refused one |
| **Every correction needs a reason, and every correction is kept** | Who changed what, when, why, and what the value was before and after. A corrected figure with no reason cannot be told apart from a figure somebody quietly changed |
| **Anybody may file a reimbursement claim, only the owner may pay one** | The person who spent the money is the right person to record it, and a supervisor may record it for somebody who cannot. Paying it spends REEF's money, so it sits with the owner alone, in the same way ordering stock does |

# 5. Recording work on the plant

Six large buttons on the worker home screen: five for the things a worker records and one for money a worker spent, which section 6 covers. Underneath them is the tile for the devices signed in to your account. Every capture screen works the same way: choose the site, fill in what you saw, save. Each takes well under a minute, because a screen slower than the paper slip it replaces will not be used.

This section is the same whether you are in a browser or in the handset application. The screens are the same screens.

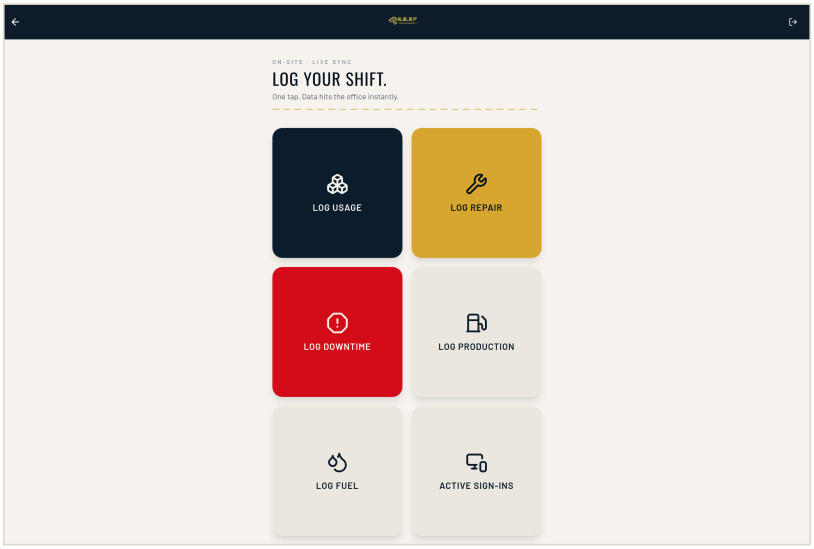



*Figure 2: The worker home screen. Nothing else is on it.*

> **NOTE**
> None of the capture screens asks you for a date. The entry is dated today, worked out in South African time rather than in the browser's time zone, so an entry captured just after midnight belongs to the new day. That is why capturing on the day matters: there is no date field to put it right with, and only a manager can change it afterwards.

> **WARNING**
> An entry cannot be dated more than sixty days ago, and it cannot be dated in the future at all. The owner sets the sixty day limit and sixty days is the default. A manager correcting a date beyond it is refused with a message saying how old the entry is. An entry that old has usually already been reported to REEF, and a report they have acted on should not change underneath them.

## 5.1 Log Production

| | |
|---|---|
| **Purpose** | Recording how many tons a site processed. |
| **Who may use it** | Worker, site manager, owner. |
| **What you should see** | A green confirmation and a return to the six buttons. The figure appears on the production register, on the dashboard and in the cost per ton for the month. |

1. From the worker home screen, select Log Production.
2. Choose the site from the Mine list.

3. Enter the figure in Tons produced.
4. Select Save Production and wait for the confirmation before closing the screen.

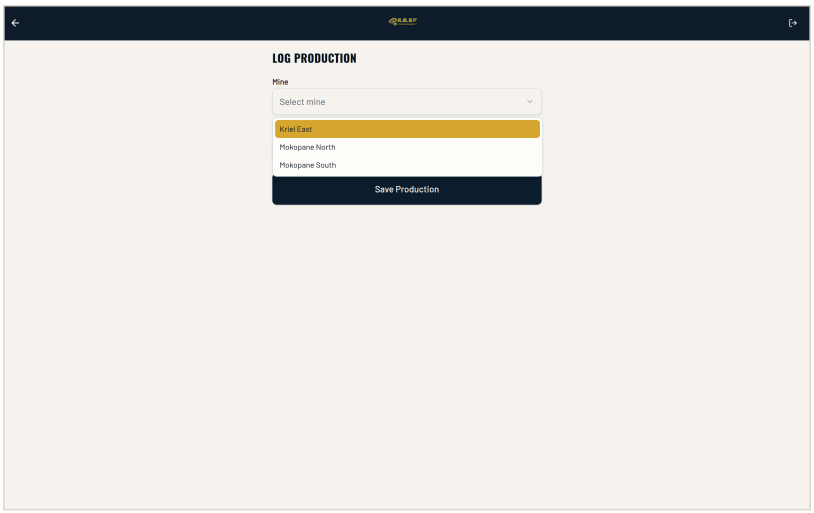



*Figure 3: Log Production, with the site list open.*

> **TIP**
> The mine list is the whole list of REEF's sites, not just yours, so read it before you tap. Tonnage saved against the wrong site moves two cost per ton figures at once, and a manager has to correct both.

## 5.2 Log Downtime

| | |
|---|---|
| **Purpose** | Recording that production stopped, for how long and why. |
| **Who may use it** | Worker, site manager, owner. |
| **What you should see** | The stoppage appears on the downtime register and in the month end report, grouped by cause. |

1. Select Log Downtime.
2. Choose the Mine.
3. Choose the Reason: Breakdown, No stock, Waiting on part, Planned maintenance or Other.
4. Choose the Equipment if one machine caused it. This is optional, and leaving it empty is right when the stoppage was not a machine.
5. Enter the Duration in hours.
6. Add Notes and Photos if they would help somebody understand it later. Both optional.
7. Select Save Downtime.

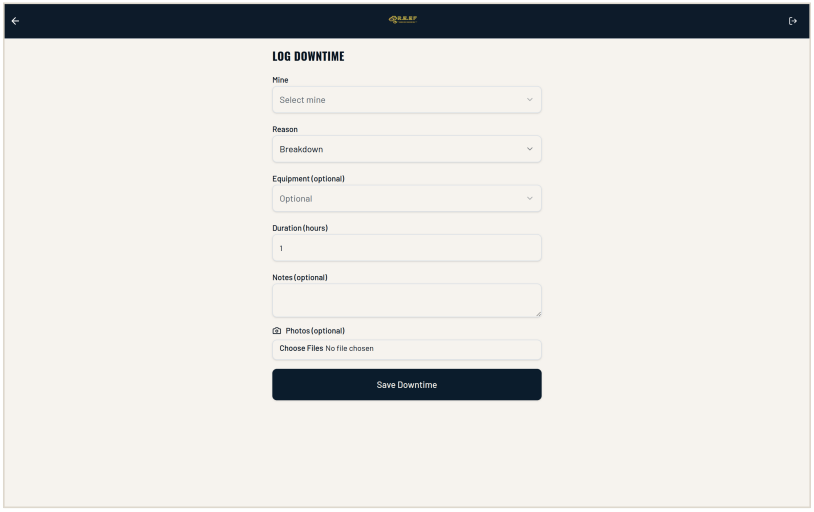



*Figure 4: Log Downtime. Reason is a list on purpose.*

> **TIP**
> Choosing a reason from the list rather than describing it in the notes is what makes downtime addable. Three people writing belt fault, belt problem and conveyor down produce three small causes in the report instead of one large one, and the real problem hides. Use Other sparingly, and say what it was in the notes.

## 5.3 Log Repair

| | |
|---|---|
| **Purpose** | Recording that a machine broke, what was done about it and what the labour cost. |
| **Who may use it** | Worker, site manager, owner. |
| **What you should see** | The repair joins that machine's maintenance history, its cost is carried into the variable side of cost per ton, and the next service falls due from it. |

1. Select Log Repair.
2. Choose the machine from the Equipment list.
3. Describe the fault in What happened, in one line and in your own words. Replaced belt on conveyor 2 is a good entry. Fixed it is not.
4. Enter the Labour cost in rand if you know it. Leave it at zero rather than guessing: a guessed cost is worse than a missing one, because nobody can tell it is a guess.
5. Add Photos if the fault is easier shown than described. Optional.
6. Select Save Repair.

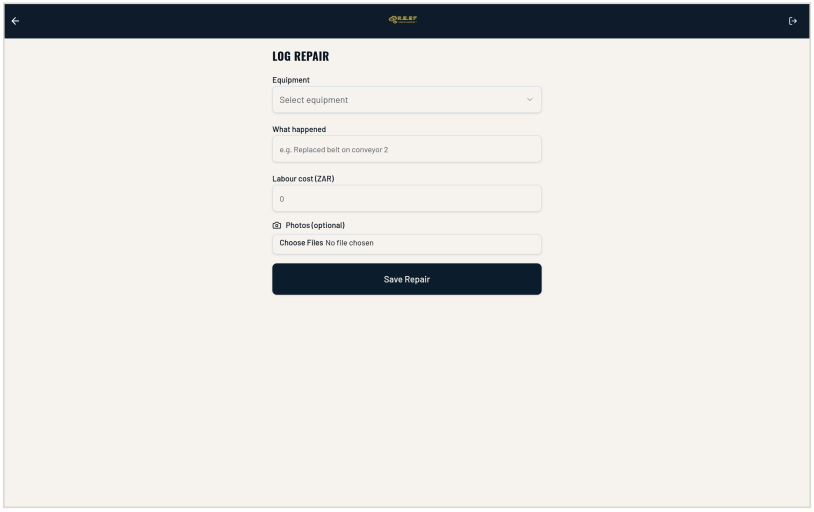



*Figure 5: Log Repair.*

> **NOTE**
> This screen records the repair and its labour. It does not take parts off the shelf. If you used parts, book them out with Log Usage, or ask your site manager to add them to the repair on the Maintenance screen, which does both at once.

## 5.4 Log Fuel Slip

| | |
|---|---|
| **Purpose** | Recording diesel drawn for a machine, so the fuel cost lands against the right site. |
| **Who may use it** | Worker, site manager, owner. Only the owner may delete one, because a slip is evidence. |
| **What you should see** | The slip appears on the fuel register and its cost is carried into cost per ton. |

1. Select Log Fuel.
2. Choose the machine in Vehicle or tool, or type a name in the box under it if the machine is not in the list.
3. Choose the Mine.
4. Enter the Litres and the Rand per litre from the slip. The total appears underneath as you type.
5. Enter the Odometer or hour reading from the machine.
6. Photograph the paper slip in Slip photo.
7. Select Save, and keep the paper slip until your manager has confirmed it.

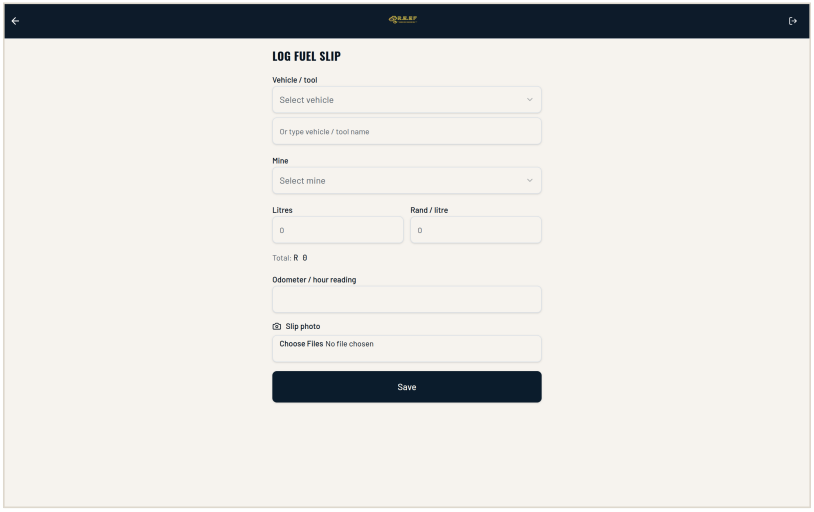



*Figure 6: Log Fuel Slip, with the running total under the two figures that make it.*

> **WARNING**
> The total is worked out here and again on the server. You cannot send a total of your own: the server refuses one and recalculates it from the litres and the price. So the two figures that matter are the litres and the rand per litre, and the price changes week to week. Typing last week's price because it is the one you remember puts a wrong fuel cost into the month.

## 5.5 Log Usage, taking a part off the shelf

| | |
|---|---|
| **Purpose** | Recording parts, consumables or tools taken out of the store. |
| **Who may use it** | Worker, site manager, owner. REEF asked that everybody be able to do this. |
| **What you should see** | The figure on the shelf drops by what you took, and the system records who took it, what, how many and when. |

1. Select Log Usage.
2. Choose the Item. The list shows each item with what the system thinks is on the shelf, and it shows only your own plant's store.
3. Check that figure against what is actually on the shelf before you go on. If the two disagree, tell your site manager rather than correcting it yourself.
4. Enter the Quantity used.
5. Select Save.

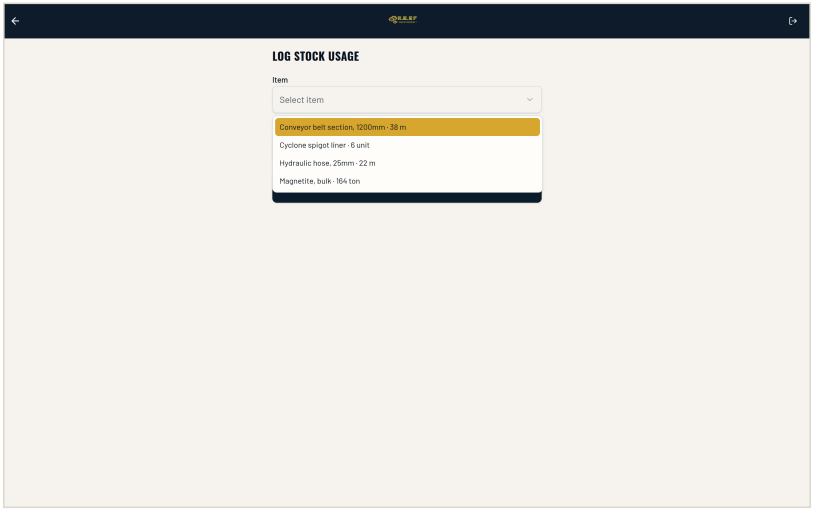



*Figure 7: Log Usage, with the item list open. Only this person's own plant is in it.*

**WARNING**

Booking a part out does not order a replacement. When the figure falls to the reorder point the system raises a reorder request, which waits for a manager. A request is not an order and nothing has been bought. If the shelf is empty, say so out loud as well as booking it out.

**NOTE**

The quantity comes off in one step inside the database. Two people booking the same part out at the same moment both count: neither reading overwrites the other. That is why the screen does not ask you for the new total, only for how many you took.

# 6. Money a worker spent, and getting it back

A worker who buys something for the site out of their own pocket records it here, and REEF pays it back at month end. On the worker screens it is the tile called Money I Spent. In the office it is the screen called Reimbursements, where the claims are read and paid.

The whole thing is deliberately short. A worker submits a claim, it waits as pending, and the owner either pays it or voids it with a reason. There is no approval step in between, because a claim that has been approved but not paid is a promise nobody is tracking.

| STATUS | WHAT IT MEANS |
|---|---|
| **Pending** | Submitted and waiting. This is where every claim starts, and where it stays until somebody acts on it |
| **Paid** | REEF has paid it back. The date it was paid and who marked it are both recorded |
| **Voided** | Not being paid, and the reason is recorded with it. A void without a reason is refused |

## 6.1 Filing a claim, as a worker

| | |
|---|---|
| **Purpose** | Recording money you paid out of your own pocket for the site, so REEF can pay it back. |
| **Who may use it** | Anybody signed in. A worker files for themselves, and a supervisor may file on somebody else's behalf. |
| **What you should see** | A confirmation reading that the claim was submitted and that you will be paid at month end, and a return to the worker home screen. |

1. From the worker home screen, select Money I Spent.
2. In What did you buy? say what it was, in one line. Two tyres from the shop in town is a good entry. Stuff is not.
3. Choose a Category: Parts, Fuel, Consumables, Tools or Other. If none of them fit, type your own in the box underneath.
4. In How much did you pay? enter the amount in rand, exactly as it is on the slip.
5. Check the Date. It starts at today. It is the date you paid, not the date you are telling the office.
6. Choose the Mine if the purchase belongs to one site. Optional, and worth doing, because it is how the office knows which site the money was spent for.
7. Add Notes if anything needs explaining. Optional.
8. Photograph the slip in Receipt photo. Optional, and the single thing most likely to get a claim paid without anybody coming back to you.
9. Select Send claim.

> **TIP**
> Photograph the slip while you still have it in your hand. You can attach more than one, up to ten, which is what to do when a single purchase came off two tills. A claim with a photograph is read once and paid; a claim without one is read, queried, and paid a week later.

NOTE

Your account has to be linked to your employee record before you can file a claim, because the claim is paid to a person rather than to a login. If you are told that this account is not linked to an employee, ask the owner to link it. It is a one off.

WARNING

A claim cannot be dated in the future and the amount has to be more than zero. Both are refused by the system rather than quietly accepted. Check the year on the date if a claim is refused and you cannot see why.

## 6.2 Reading the claims, as a manager or the owner

| | |
|---|---|
| **Purpose** | Seeing what is owed, checking each claim against its receipt, and paying them. |
| **Who may use it** | A worker sees their own claims. A site manager sees the claims at their own plant. The owner sees all of them, and the owner is the only person who can mark one paid or void it. |
| **What you should see** | A list with the newest claim first, and the total still outstanding across however many claims are pending. |

1. From the menu, select Reimbursements.
2. Use the tabs to choose what you are looking at: pending, paid, voided or all. It opens on pending, because that is the list that needs a person.
3. Use the month box to narrow it to one month, and Clear month to come back out.
4. Read the figure on the right. It is the total outstanding across the pending claims shown.
5. Select View on a row to open one claim: the worker, the amount, the category, the date, the notes and the receipt photograph.

NOTE

The separation between plants applies here too. A site manager sees the claims of workers at their own plant and nobody else's, and that is enforced in the database rather than by hiding rows on the screen. A claim with no plant recorded against it reaches the owner alone.

## 6.3 Paying the claims at month end

This is the owner's job, and it is built to be done in one sitting rather than one claim at a time.

1. Open Reimbursements and stay on the pending tab.
2. Tick the claims you are paying. The box in the heading row ticks every pending claim at once, which is the usual month end case.
3. Select Mark N as paid.
4. Read the confirmation before you agree to it. It lists every claim you ticked, the total, and the total broken down by worker, which is the figure you are actually paying each person.
5. Select Confirm to pay. Every claim in the batch is marked in one step, so a payout cannot end up half done.

**WARNING**

Marking a claim paid is a record that REEF has paid it, not an instruction to a bank. The system does not move money. Pay the people the way REEF normally pays them, and mark the claims paid once that is done, not before.

**NOTE**

To refuse a claim, open it and void it with a reason. The reason is required and it is kept, so a worker whose claim was not paid can be told why rather than being left to ask. Every change to a claim is written to the change history in section 11.3 in the same step as the change itself.

## 6.4 What a reimbursement does not do yet

A reimbursement is a record of money REEF owes a worker and then paid. It does not yet flow into the cost per ton figures in section 12, and it does not add a part to the stock figure even when the purchase was a part. If a worker bought something that is now on the shelf, book it in on Inventory as well as claiming for it, and say in the notes that you have done so.

# 7. The handset application

REEF now has an application a worker installs on an Android phone. It is the same system you reach in a browser, not a different one: the same sign in, the same buttons, the same rules underneath. What it changes is how a worker gets to it. An application has its own icon on the home screen and opens with one tap, and a worker in gloves at the start of a shift does not have to find a browser and type an address.

## 7.1 What it is, and what it is not

| | WHAT THAT MEANS FOR YOU |
|---|---|
| **It is an application** | Its own icon, its own name on the phone, its own splash screen. It is installed, not bookmarked |
| **It is the same system** | It opens the deployed REEF system inside itself, so a fix or a new screen reaches the handset at the same moment it reaches the office. The phone can never be a version behind |
| **It grants nothing new** | Same email and password, same role, same plant, same refusals. Installing it gives a worker nothing the browser did not already give them |
| **It is Android only in this release** | An iPhone reaches the system in Safari, which works and is covered everywhere else in this manual |
| **It still needs a connection** | It cannot hold a shift's work while you are out of signal. Section 7.5 says what it does instead and what is coming |

## 7.2 Installing it

The application is not in the Play Store in this release. Your site manager has the installation file, which is called an APK, and gives it to you directly.

1. Ask your site manager for the REEF installation file. They will send it to you or put it on the phone for you.
2. Open it on the phone. Android will warn you that it did not come from the Play Store. That warning is expected for an application a company installs on its own phones.
3. Allow the installation when the phone asks, and wait for it to finish.
4. Find REEF on the home screen and open it.
5. Sign in with the same email address and password you use in the browser. If you have never signed in before, ask your site manager to have an account created for you first.

> **NOTE**
> You only do this once. After that the application updates itself without being reinstalled, because it opens the deployed system rather than carrying its own copy of the screens. A new screen appears the next time you open it.

## 7.3 Using it

Everything from here is sections 5 and 6. The worker home screen, the buttons, the fields and the confirmations are the same. There is nothing extra to learn, which was the point of building it this way.

Two small things are worth knowing. The buttons and fields were sized for a thumb in a glove before there was an application to put them in, so nothing in it is smaller than the Android minimum target size. And signing out is in the same place it is in the browser, at the top right, which matters on a phone that is shared between shifts.

## 7.4 If the phone is lost

Exactly what section 15 says, and it is worth knowing that installing the application does not change it. Tell your site manager at once. The owner signs that device out from the Active Sign-ins screen, and it stops working on its very next action. Your password does not have to change, and nothing useful is left on the phone, because the application does not keep your records on it.

## 7.5 What it does not do yet

It cannot capture without a connection. With no signal it shows a plain page saying you are offline, with one button to try again. That is deliberate: the alternative is a screen that looks like it saved and did not, and a worker who finds out a week later that a shift is missing.

Holding a shift on the handset until signal returns is designed and is the next thing being built. When it arrives, an entry you save with no signal will be kept on the phone, encrypted, and marked Queued rather than confirmed. It will survive a locked phone and a restart. It will send itself the moment the phone is back in coverage, and only once REEF has accepted it will it read Sent. Until then you will be able to see exactly what is still owed. None of that is in this release.

> **WARNING**
> Until that arrives, treat a part of the plant with no coverage the way you did before: write the figures down, and enter them when you are back in signal, the same day. The sixty day rule in section 5 is not a problem for a same day entry, but the entry is dated the day you type it, so a figure from yesterday typed today needs a manager to correct its date.

# 8. The dashboard

Where a site manager and the owner land. One screen answering how things are going, and three panels naming what is waiting on a person.

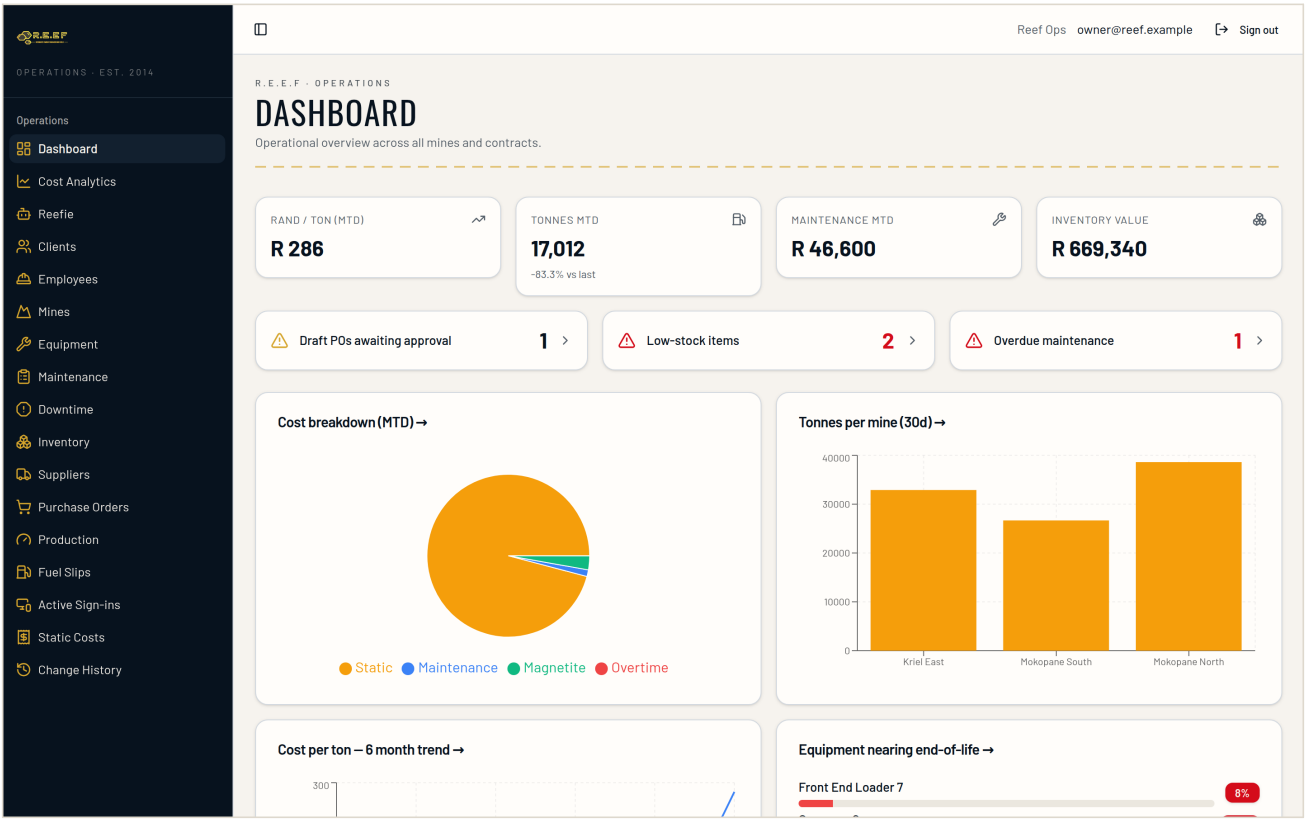



*Figure 8: The dashboard, seen by the owner. A site manager sees the same screen with their own plant's figures in it.*

## 8.1 The four figures at the top

| FIGURE | WHAT IT MEANS |
|---|---|
| **Rand per ton (MTD)** | Everything the business has spent this month divided by the tons produced this month. MTD is month to date, so on the fifth of the month it covers five days |
| **Tonnes MTD** | Tons produced this month, with the change against the same point last month underneath |
| **Maintenance MTD** | What repairs have cost this month, labour and parts together |
| **Inventory value** | What is on the shelf, valued at each item's unit cost. Every plant for the owner, your own for a manager |

> **NOTE**
> Rand per ton looks high in the first days of a month. Salaries, leases and insurance are entered as whole months, and in the first week they are divided by a few days of tonnage. It settles as the month fills. The month end report in section 12 apportions fixed costs by days and does not have this effect.

## 8.2 The three panels, and what to do about each

| PANEL | WHAT IT IS TELLING YOU | WHAT TO DO |
|---|---|---|
| **Draft POs awaiting approval** | Orders the system drafted, or a manager created, that nobody has approved. Nothing has been bought | Open Purchase Orders, check the lines against what the plant needs, and approve or cancel. A draft nobody looks at is a part nobody ordered |
| **Low stock items** | Items at or below their reorder point, counted for the plants you can see | Open Inventory, check the shelf, and raise or approve an order |
| **Overdue maintenance** | Machines whose next service date has passed | Open Maintenance and book the service in. A reminder has also gone to your notifications |

Each panel is a button. Selecting one opens the list behind it, and selecting a row in that list opens the record.

## 8.3 The four charts

| CHART | WHAT IT SHOWS |
|---|---|
| **Cost breakdown (MTD)** | This month's spend split into static, maintenance, magnetite and overtime. The slice that grew is the thing to ask about |
| **Tonnes per mine (30d)** | The last thirty days of production, site by site |
| **Cost per ton, six month trend** | Rand per ton for each of the last six months. A month with no production has no point on the line rather than a zero |
| **Equipment nearing end of life** | The five machines with the least life left, measured in tons against what they were expected to carry |

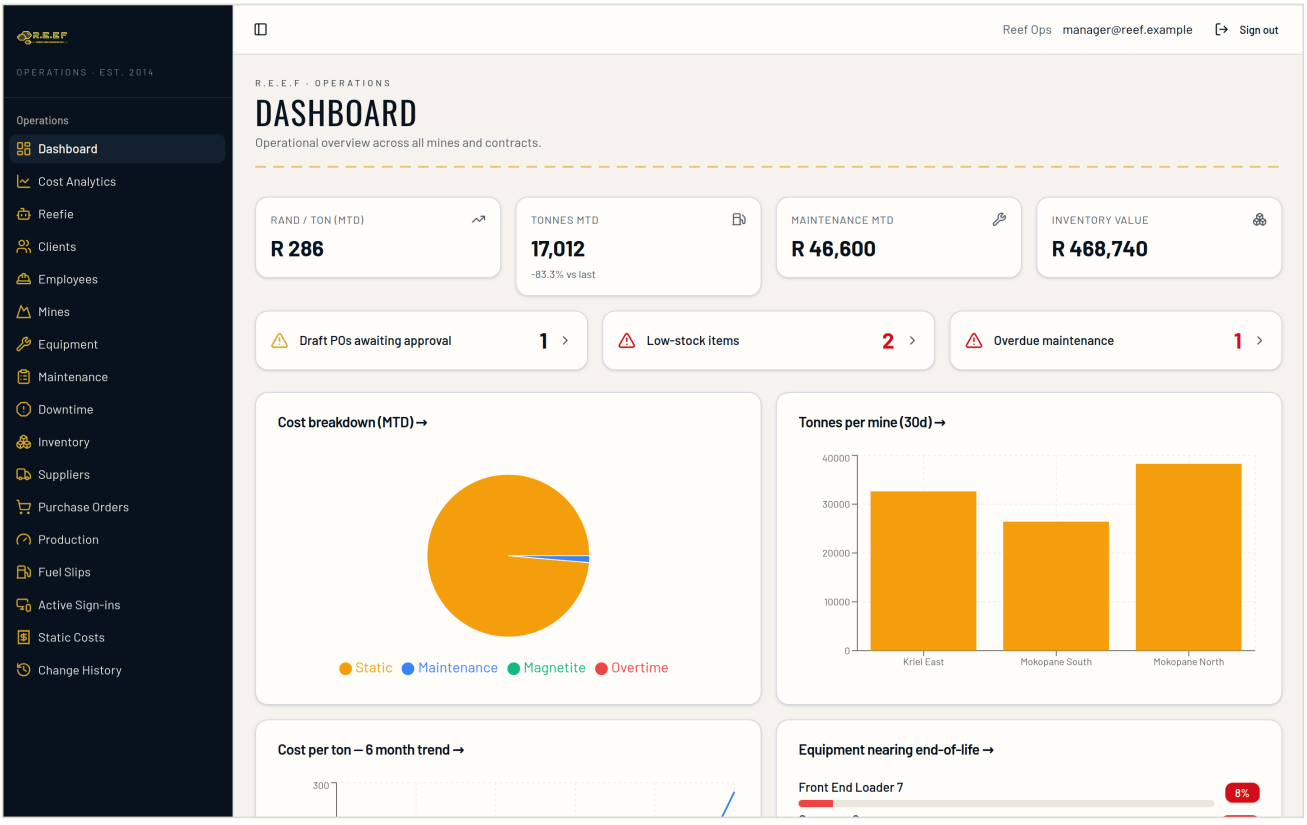



*Figure 9: The same dashboard as a site manager at Mokopane Plant 1. The inventory value and the low stock count cover that plant alone.*

# 9. Stock and purchasing

Stock is the part of the system REEF's written answer changed most. Each plant keeps its own count of the same part, the two counts are never added together, and a plant does not see the other plant's shelf. The owner is the only person who sees both.

> **NOTE**
> If a part shows twenty in stock, that is twenty at your plant. The other plant keeps its own figure for the same part. They are two numbers, not one shared one, and ordering against the wrong one is what this separation prevents.

## 9.1 How a part gets from the shelf to a delivery

| | STEP | WHO DOES IT | WHAT IT IS |
|---|---|---|---|
| 1 | A part is booked out | Anybody, on Log Usage or through a repair | The figure on the shelf drops |
| 2 | A reorder request is raised | The system, by itself | Raised the moment the figure reaches the reorder point. A request is not an order |
| 3 | The request becomes a draft order | A site manager or the owner | This is the step that commits the request to the purchasing workflow |
| 4 | The draft is approved | A site manager or the owner | This is the step that spends REEF's money |
| 5 | The order is marked sent | A site manager or the owner | Optional. An order can be received straight from approved |
| 6 | The delivery is marked received | A site manager or the owner | This is the step that raises the stock figure. It can only happen once |

## 9.2 Inventory

| | |
|---|---|
| **Purpose** | The parts and consumables at a plant, what is on the shelf, and the level at which more should be ordered. |
| **Who may use it** | Everybody may read it. Only a site manager or the owner may add an item or change a figure. |
| **What you should see** | A list with anything at or below its reorder point marked Low in red. |

1. From the menu, select Inventory.
2. Read the On hand column against Reorder at. The two figures in that column are the reorder point and the quantity to order when it is reached.
3. As the owner, use the Plant list at the top to look at one plant at a time. A manager has no such list, because they have one plant.
4. Select New stock item to add a part, or the pencil on a row to change one.

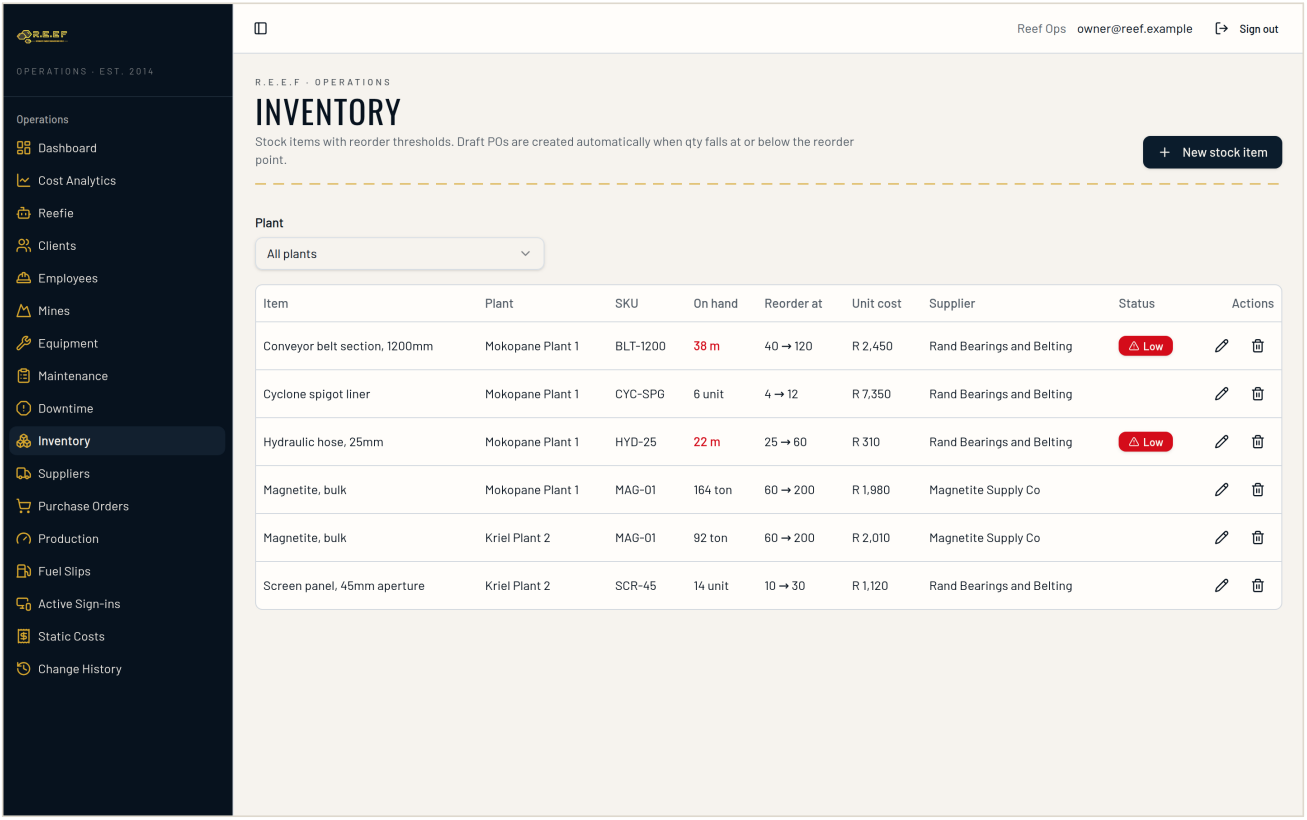



*Figure 10: Inventory, seen by the owner. The Plant column and the Plant filter appear for the owner only.*

> **NOTE**
> An item and its levels are two records. The item is the part itself, its name, its code, its unit, its cost and its supplier. The level is how many of them are at one plant and when to reorder. That is what lets the same part exist at both plants with two separate counts.

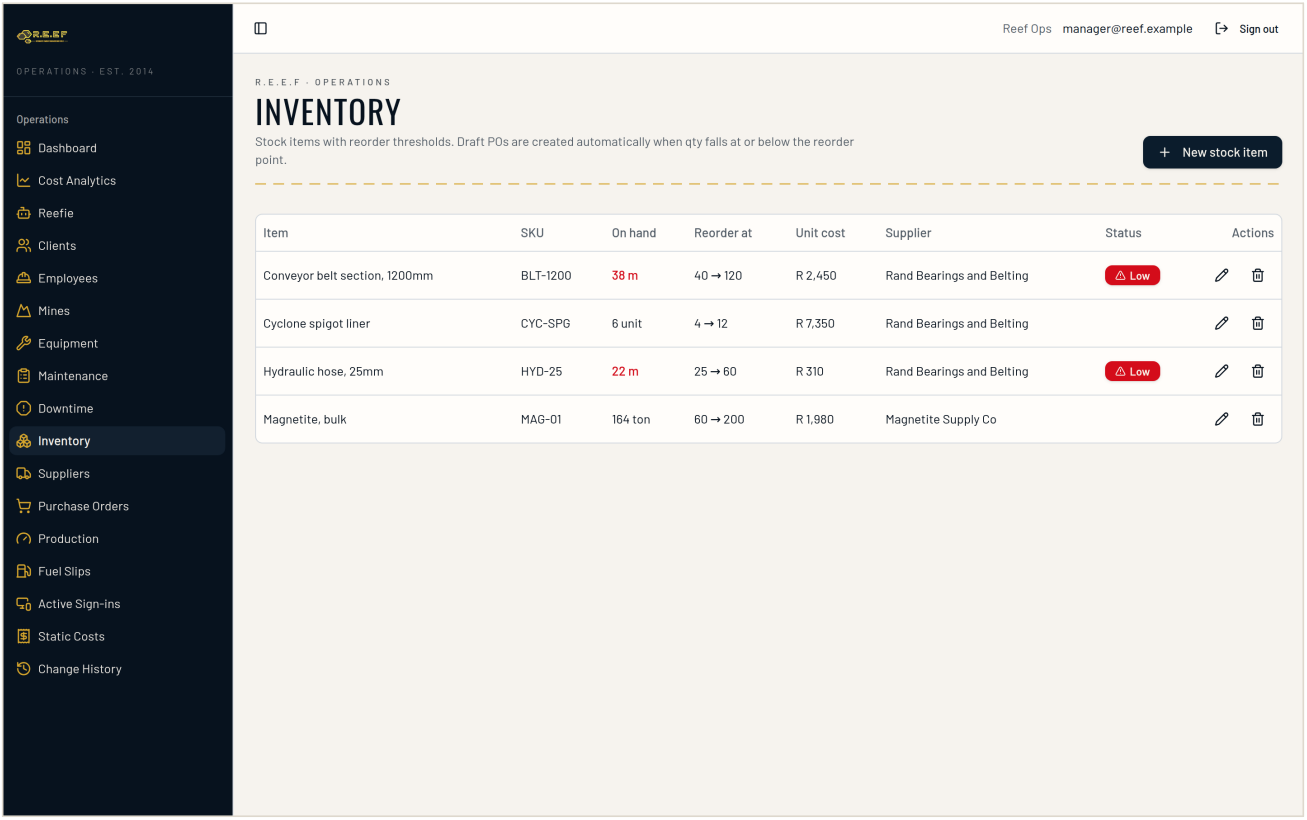



*Figure 11: The same screen as a site manager. One plant, no Plant column, no filter, and no way to ask for another plant's shelf.*

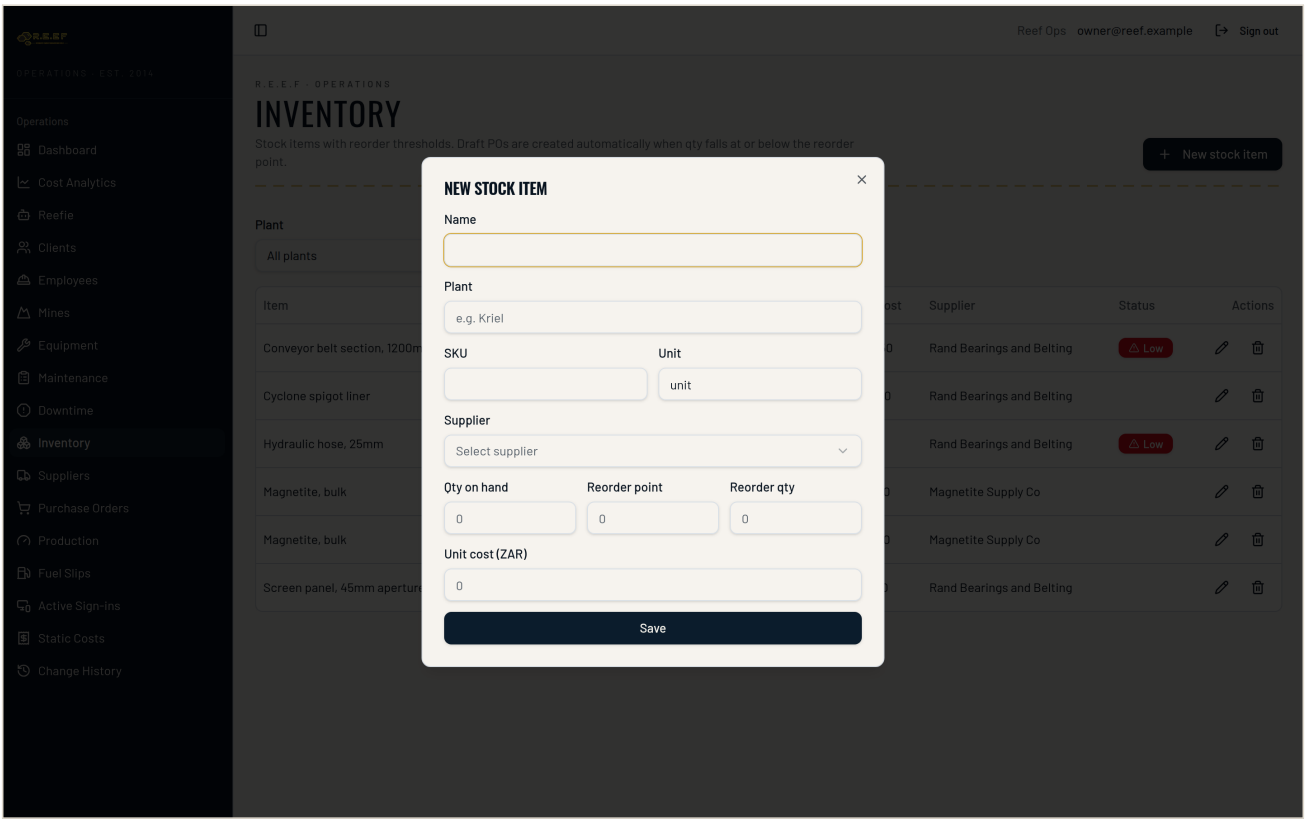



*Figure 12: Adding a stock item. The owner names the plant; for anyone else the plant is their own and the field is not shown.*

## 9.3 Purchase Orders

| | |
|---|---|
| **Purpose** | Ordering from a supplier, approving an order, and recording that it arrived. |
| **Who may use it** | Site manager and owner only. A worker cannot reach this screen at all. |
| **What you should see** | A list of orders with their status, and stock figures that rise when an order is marked received. |

1. From the menu, select Purchase Orders.
2. Read the Status column. A draft has not been ordered. approved has been committed to. ordered has been sent to the supplier. received is on the shelf.
3. Select View on a row to see its lines.
4. Select Approve to commit to the order, then Mark ordered when it goes to the supplier.
5. When the delivery arrives and has been counted, select Mark received.
6. Select Cancel PO to stop an order that has not arrived. A reason is required.

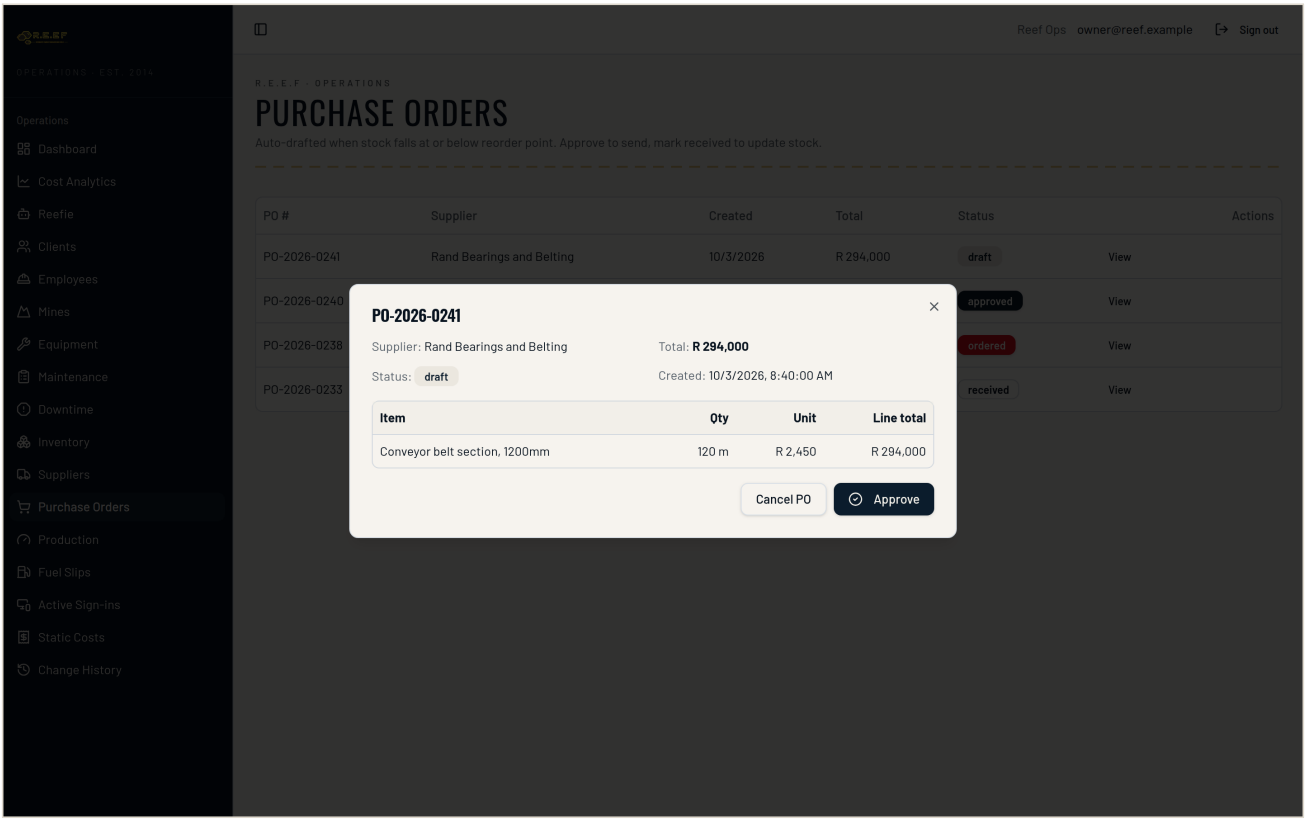



*Figure 13: A draft order opened. The lines, the total, and the two things that can be done to it.*

**WARNING**

Mark received only after the delivery has been counted, not when it arrives. Marking it received is what raises the stock figure, and an order can only be received once, so it cannot be corrected by doing it again. Receiving on the strength of a delivery note puts parts on the shelf in the system that are not on the shelf in the store.

**NOTE**

The order of those steps is enforced in the database, not on the screen. Lines can only be added to a draft, the status only moves forward, and a received order cannot be received again. Those rules therefore hold whatever is used to reach the record.

## 9.4 Suppliers

| | |
|---|---|
| **Purpose** | Who REEF buys from, and how to reach them. |
| **Who may use it** | Site manager, owner. |
| **What you should see** | A list the Inventory and Purchase Orders screens choose from. |

1. From the menu, select Suppliers.
2. Select New supplier, or the pencil on a row.
3. Keep one record per supplier. Two records for the same company split its orders in two and make the spend look smaller than it is.

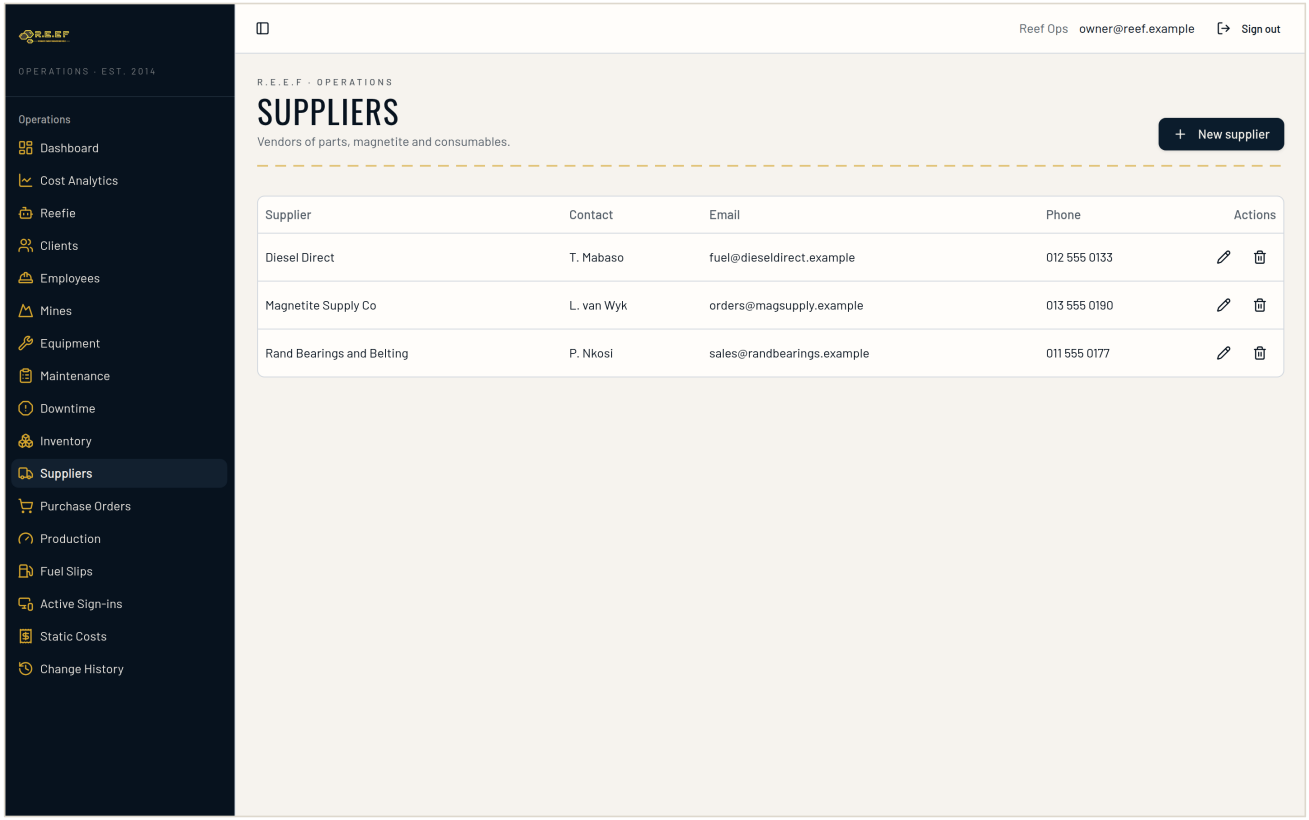



*Figure 14: Suppliers.*

# 10. The registers

Nine screens, one per kind of record. They all behave the same way: a list you can sort by selecting a column heading, a button to add a record, and a pencil and a bin on every row. Most carry a chart or a few summary figures above the list. Every one of them has a search box, which also tells you how many rows matched and has a Clear button once you have typed something.

| REGISTER | WHAT IT HOLDS | WHO MAY CHANGE IT |
|---|---|---|
| **Mines and Sites** | The sites REEF operates, their client, their team and the cost per ton each one is aiming at | Manager, owner |
| **Equipment** | Every machine and major component, its site, its install date, its expected life and its service interval | Manager, owner |
| **Maintenance and Repairs** | Every repair with its labour, its parts and its total, and when the next service falls due | Manager, owner |
| **Downtime** | Every stoppage with its cause and its hours, and a chart of hours lost by cause | Manager, owner |
| **Production** | Tonnage by site, day and shift, with magnetite and overtime | Manager, owner |
| **Fuel Slips** | Litres, price, total and odometer reading, with the photograph of the slip | Manager to correct, owner to delete |
| **Clients** | The mining companies REEF holds contracts with | Manager, owner |
| **Employees** | The crew, their site, shift, team and rate, with attendance and transfers | Manager, owner |
| **Static Costs** | The monthly fixed costs: salaries, leases, insurance and licences | Manager, owner |

**WARNING**

Static Costs is the one register nothing fills in for you. Salaries, rent, insurance and licences do not arrive from a capture screen. Cost per ton is wrong every month nobody enters them, and wrong in the direction that makes the plant look cheaper than it is. One row per month per category, against a site or marked company wide.

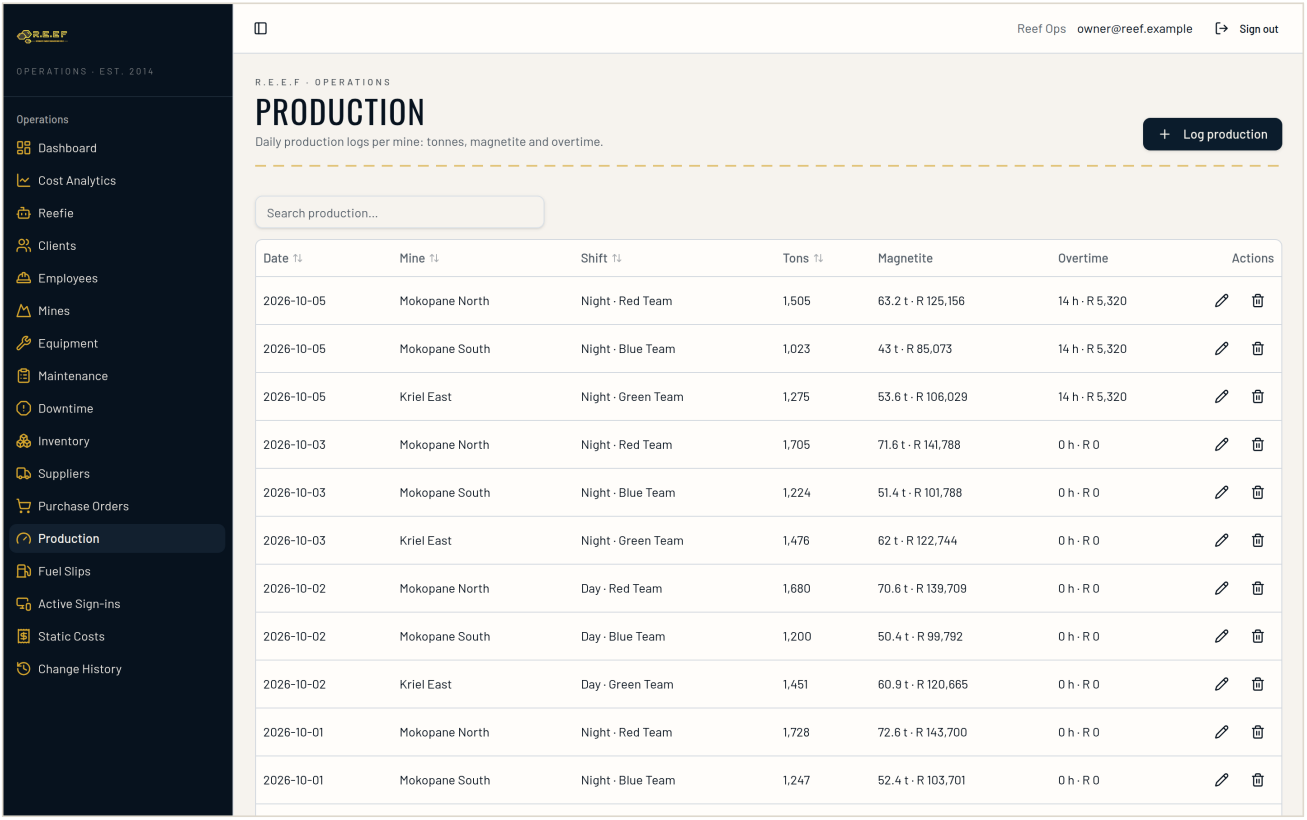



*Figure 15: Production. The summary figures above the list cover what is shown.*

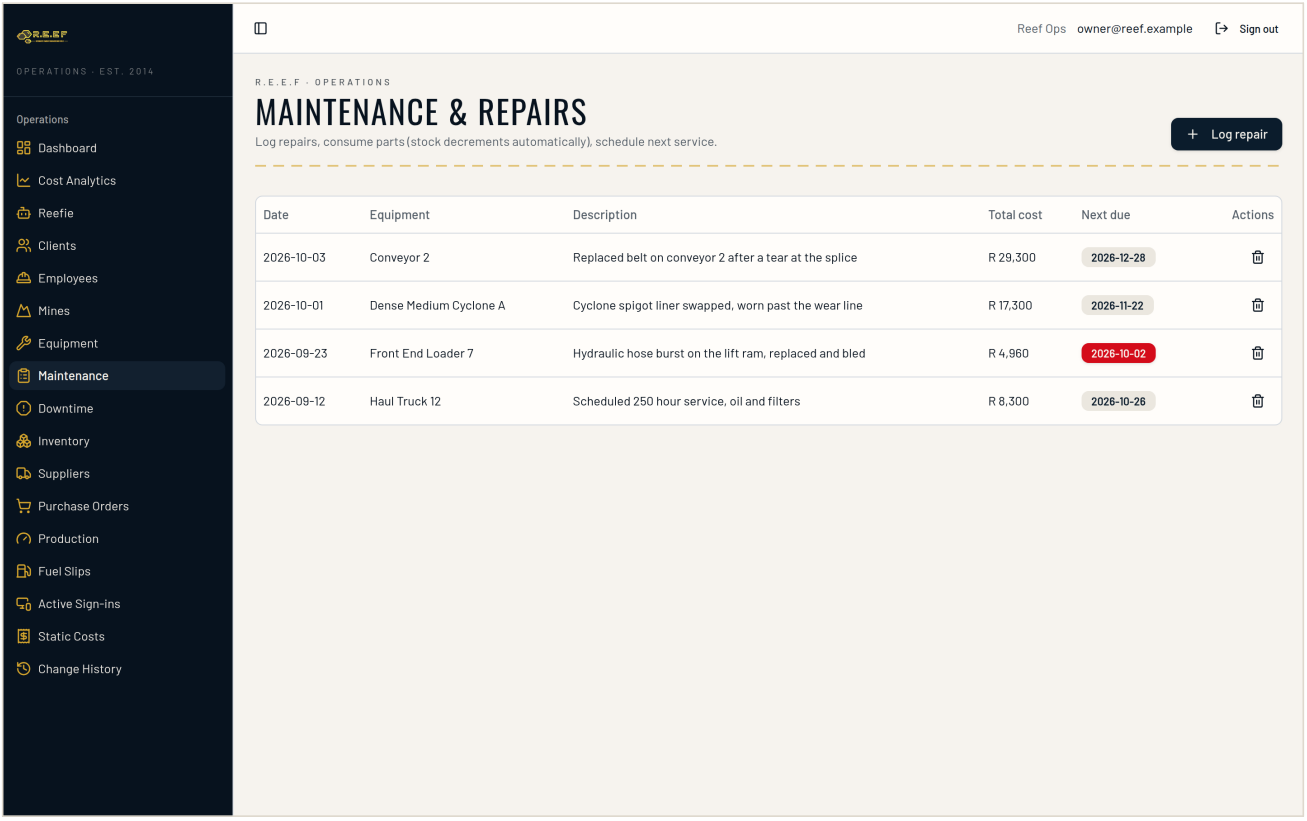



*Figure 16: Maintenance and Repairs. Parts are added to a repair from here, which takes them off the shelf in the same step.*

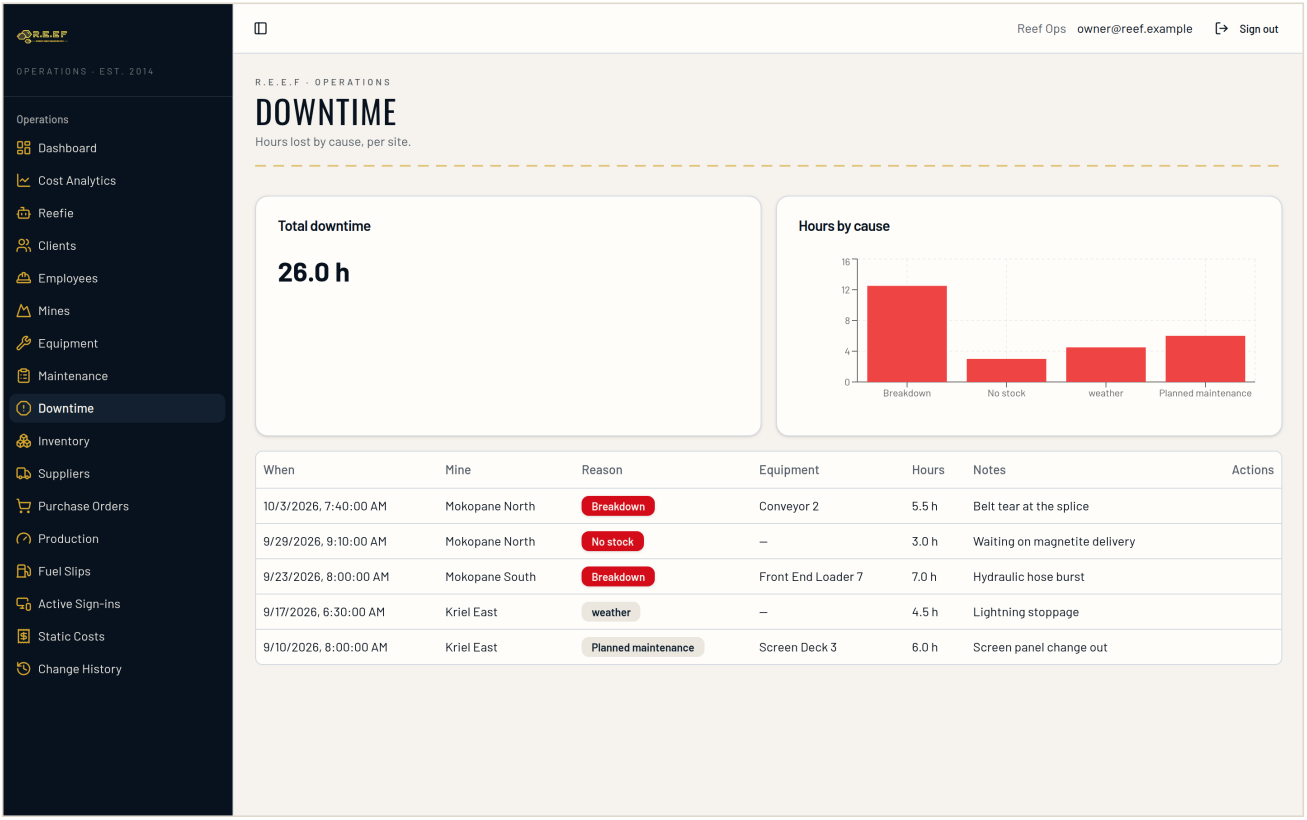



*Figure 17: Downtime, with hours lost by cause. This is the chart that makes the case for choosing a reason from the list.*

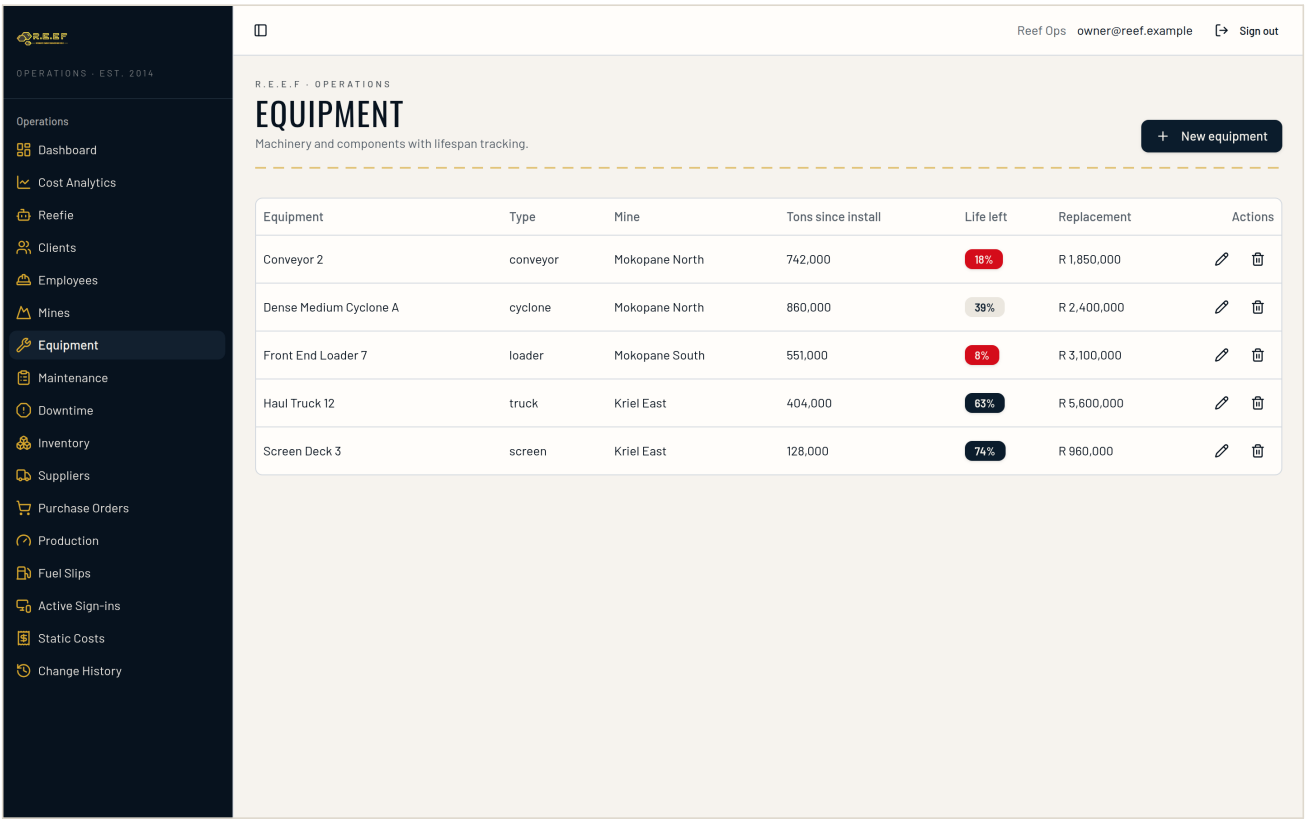



*Figure 18: Equipment. The life remaining column is what the dashboard's end of life panel reads.*

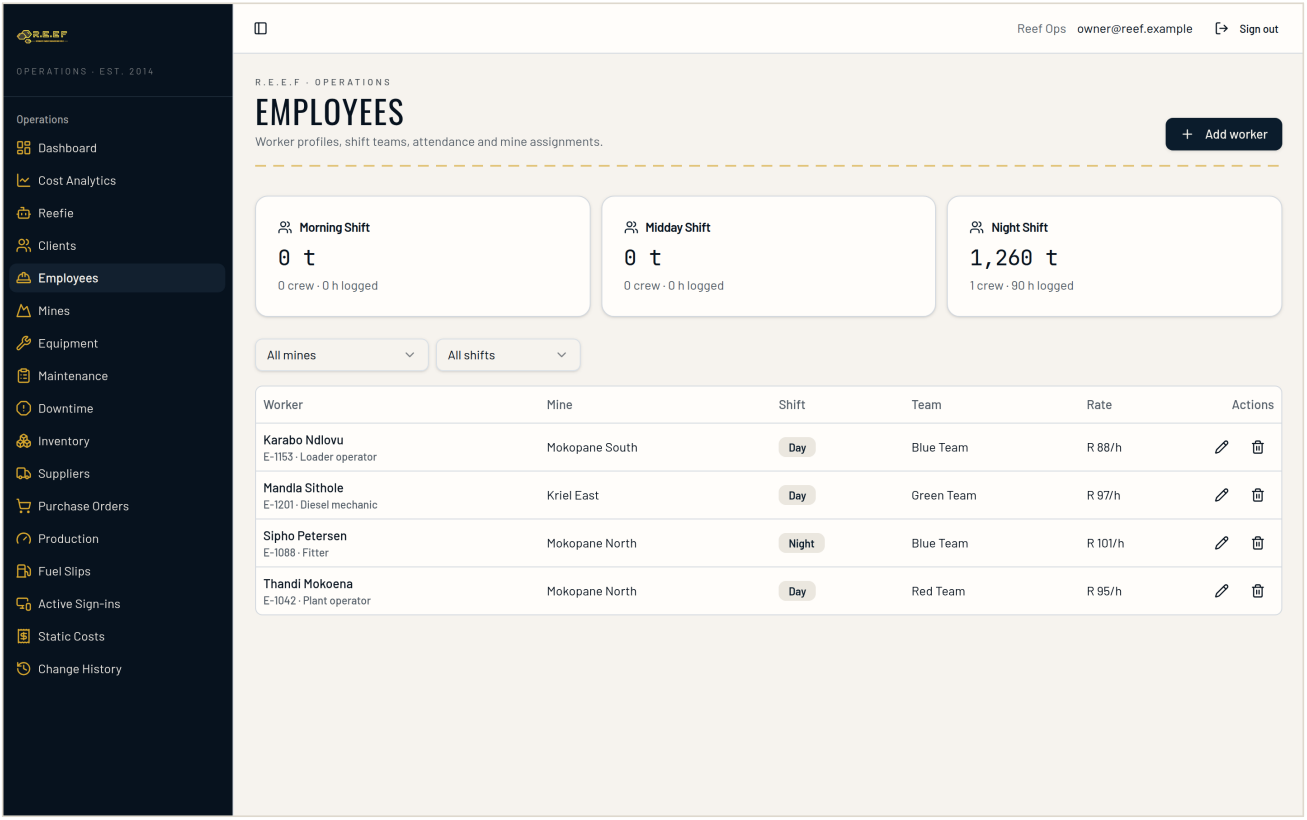



*Figure 19: Employees.*

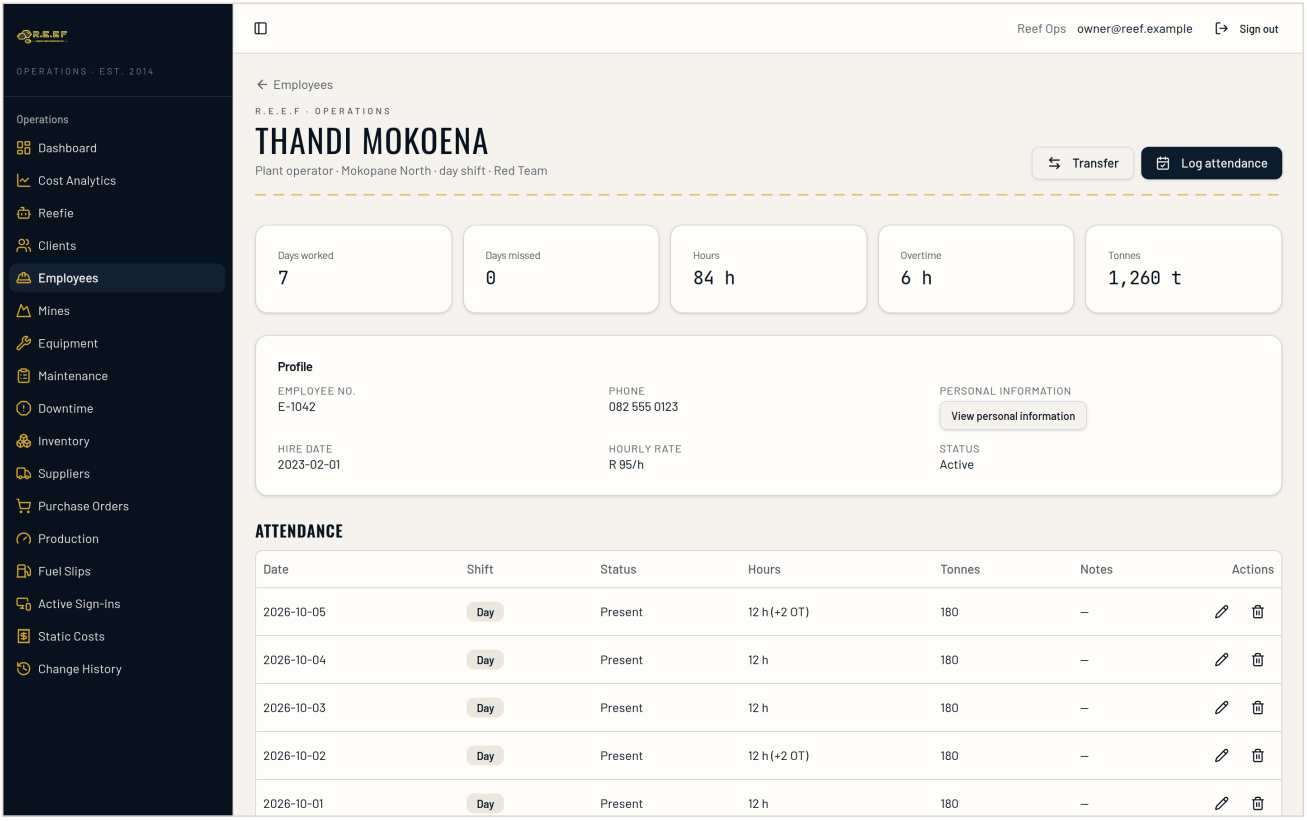



*Figure 20: One employee, with attendance and transfers. View personal information is owner only, and every use of it is recorded.*

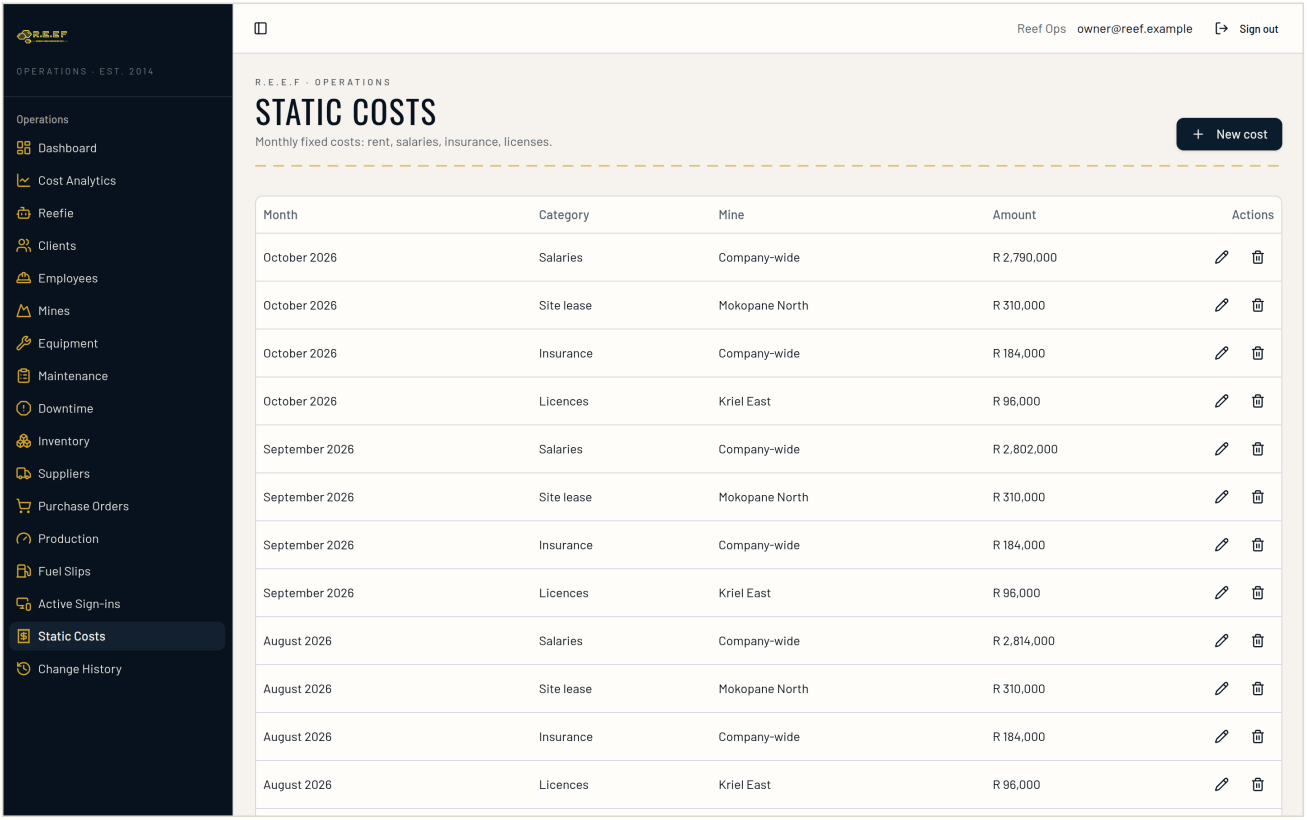



*Figure 21: Static Costs. Company wide rows are shared across sites by day when the month end report is produced.*

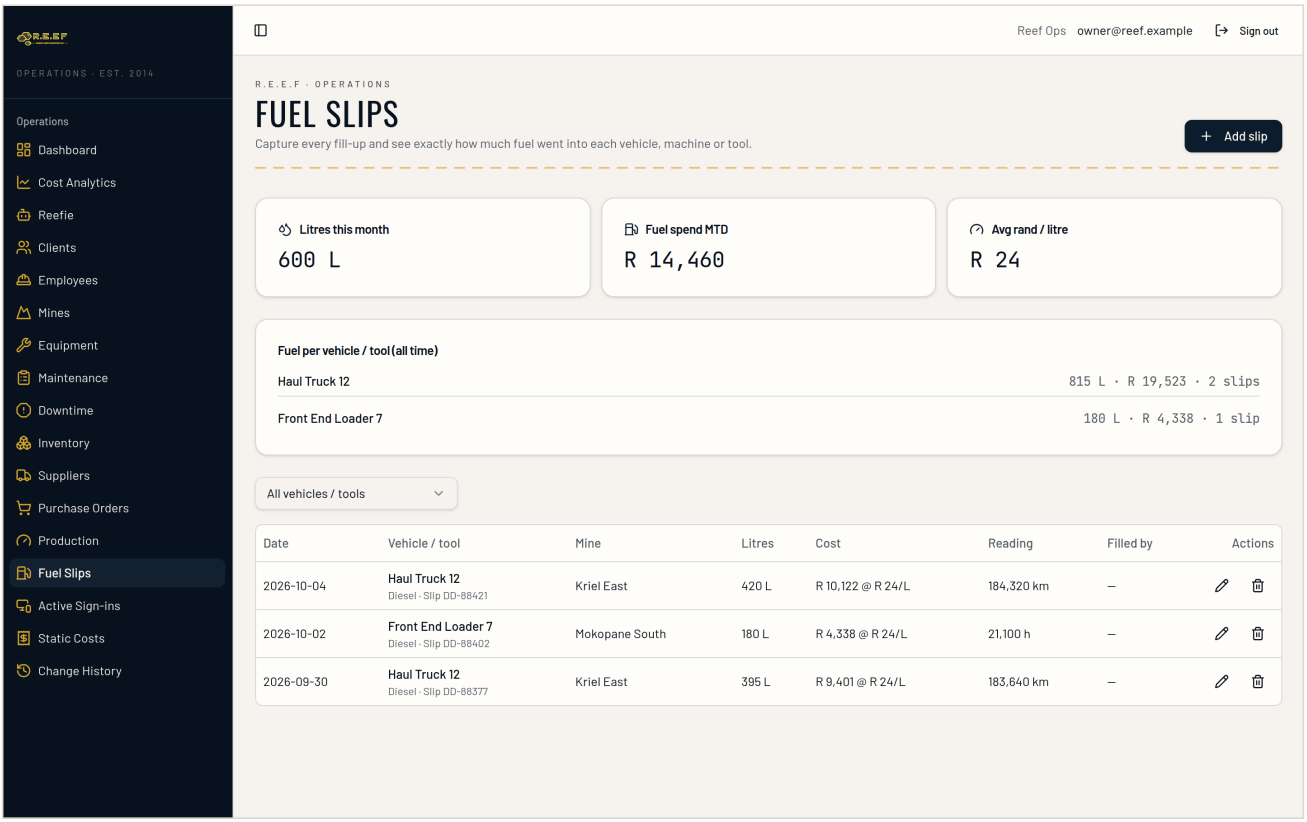



*Figure 22: Fuel Slips.*

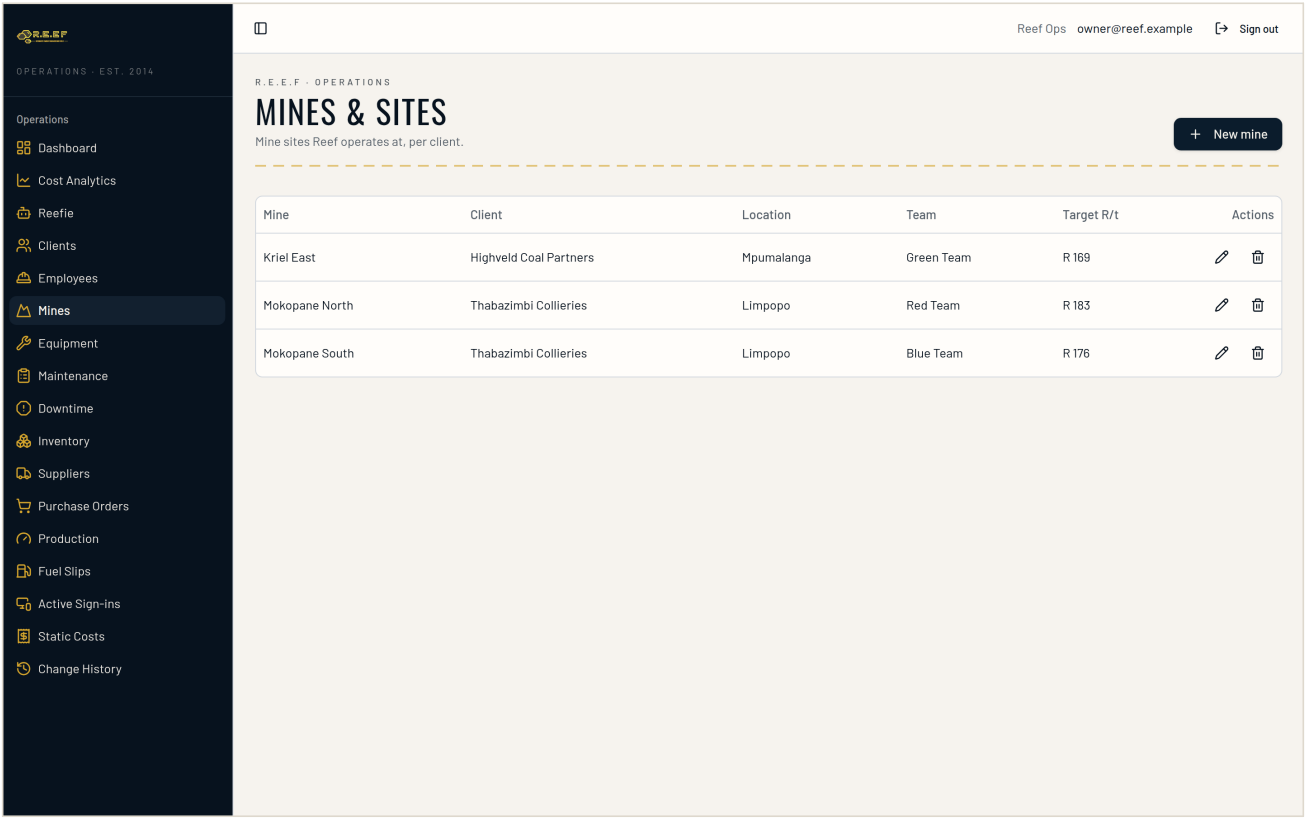



*Figure 23: Mines and Sites. The target rand per ton on each row is what the cost figures are measured against.*

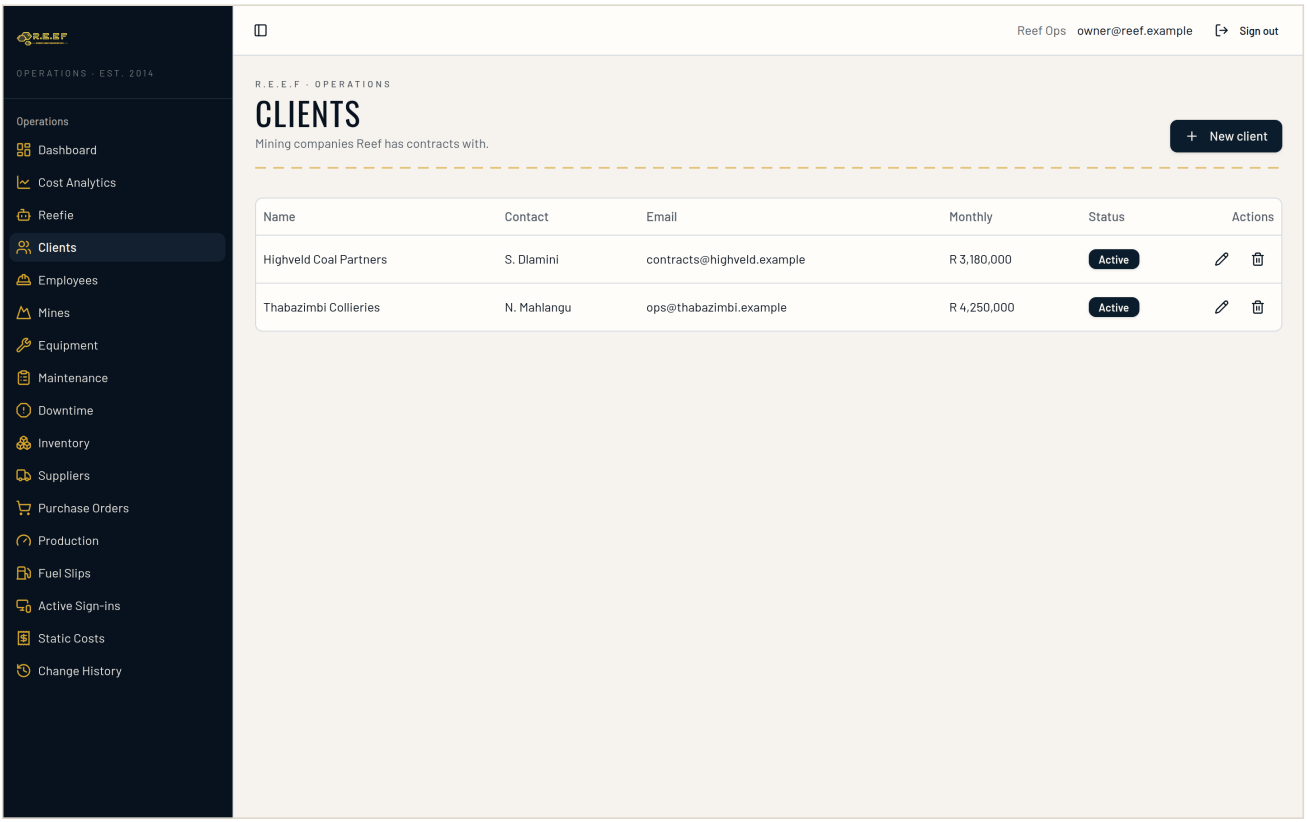



*Figure 24: Clients.*

# 11. Correcting a record

Anything captured can be put right. What the system insists on is that a correction says why, that it cannot quietly overwrite somebody else's, and that it is kept.

## 11.1 Making a correction

1. Open the register the record is on and select the pencil on its row.
2. Change only what is wrong. Leave the rest alone.
3. Type the reason for this change. One line is enough, as long as it says why rather than what. Weighbridge reprint, the first figure was the gross is a reason. Corrected is not.
4. Select Save.

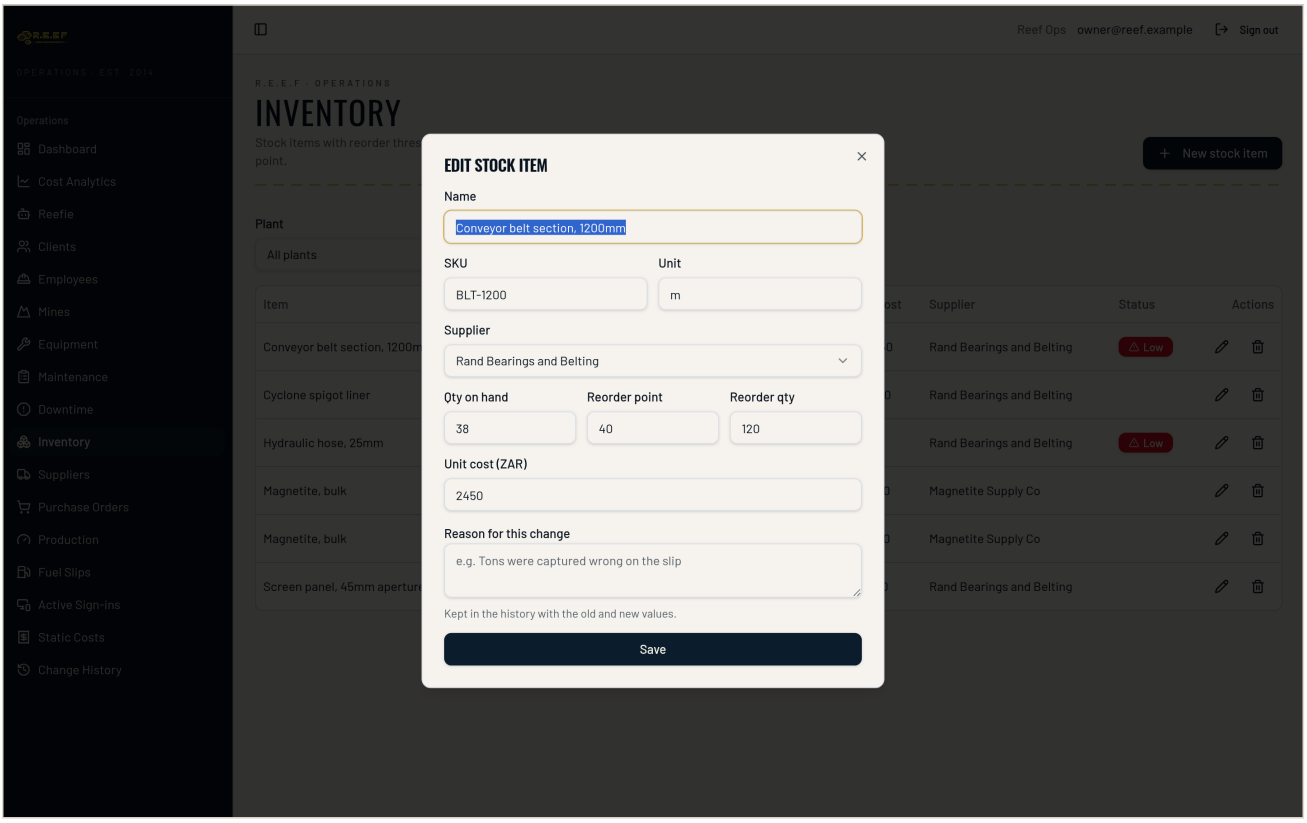



*Figure 25: An edit dialog. The reason box is on every one of them, and the save is refused without it.*

> **NOTE**
> Adding a record does not ask for a reason; changing one does. A new record has nothing to explain. A changed one does, because a figure that moved after it was first written is the kind of thing somebody will ask about.

## 11.2 When somebody else saved first

Two people can have the same record open. If the other person saves while you are typing, your save is refused rather than allowed to overwrite theirs. The dialog stays open, keeps everything you typed, and shows you their version next to yours, field by field.

1. Read what changed. The notice lists only the fields that differ.

2. Decide whether your change is still needed. Often it is not, and you can close the dialog.
3. If it is, select Load their version and keep my typing. The fields you changed keep your typing; the fields you did not touch take theirs, so saving again cannot undo a change you never saw.
4. Select Save again.

> **NOTE**
> Most systems let the second save win silently, and the person whose change vanished never finds out. Here neither change is lost, because the second save is refused until the person has seen the first.

## 11.3 Change History

Every change to a saved record is kept: which record, who changed it, when, the reason they gave, and the values before and after. The list shows the latest two hundred, newest first, and can be narrowed to one kind of record or searched.

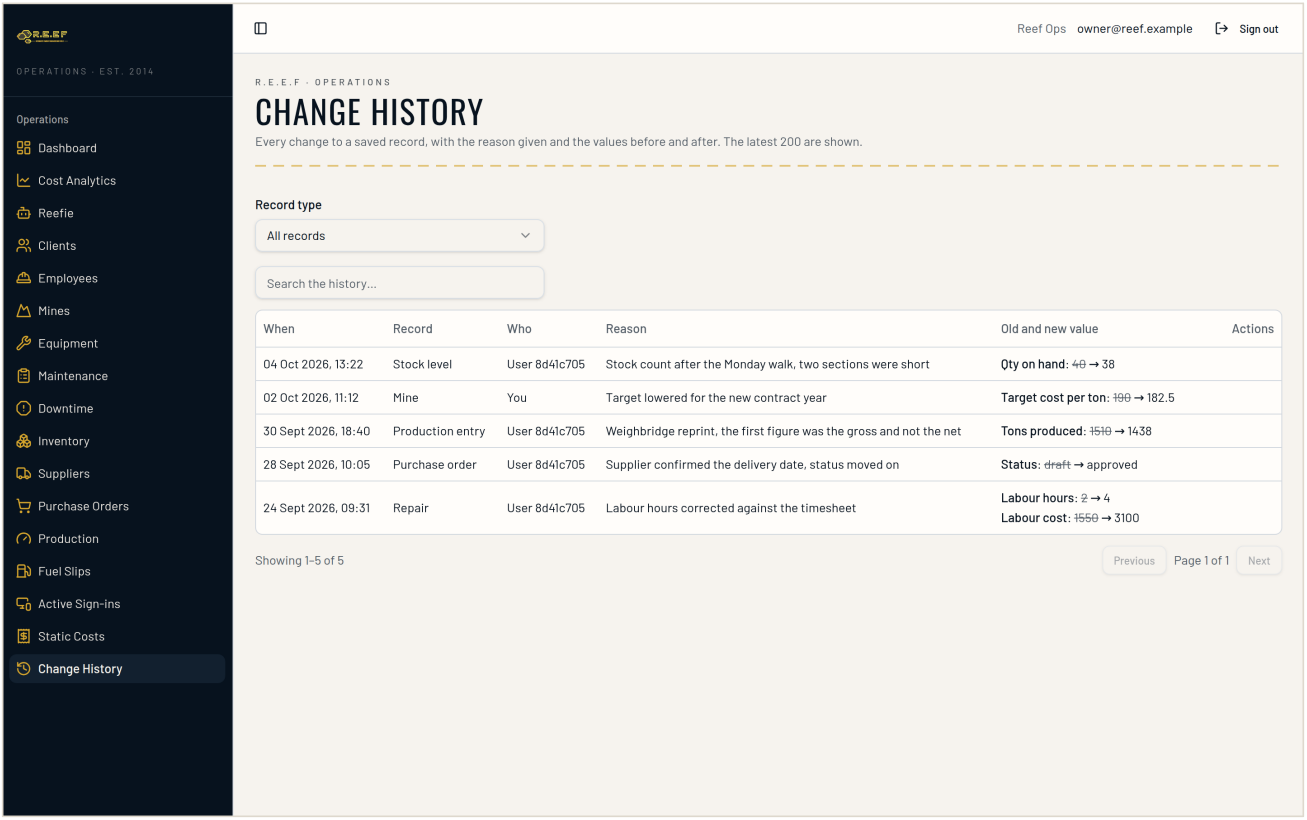



*Figure 26: Change History. The old value is struck through and the new one follows it.*

| WHO MAY READ IT | WHAT THEY SEE |
|---|---|
| **Owner** | Every change, at every plant |
| **Site manager** | Changes to records at their own plant |
| **Worker** | Nothing. The history can show pay and personal details |

NOTE

Two kinds of entry appear without a reason, and both are honest about it. Automatic means another change caused this one: logging a part took it off stock, receiving an order put stock on. Not given means the record was changed outside the usual screens. Neither can be hidden, because the history is written by the database in the same step as the change itself.

# 12. Cost per ton, and the month end report

This is what REEF asked for first. Everything a site spends, divided by the tons it produced, with the fixed and the variable costs kept apart so a rise can be explained rather than only noticed.

## 12.1 Cost Analytics

1. From the menu, select Cost Analytics.
2. Read Blended rand per ton first: every site together over the last twelve months.
3. Read Variable share next to it. It says how much of the spend moves with production and how much does not. A plant with a high fixed share is one that suffers badly in a slow month.
4. Use Cost composition by month to see which part of the cost moved: static, maintenance, magnetite or overtime.
5. Use the table at the bottom for one site at a time.

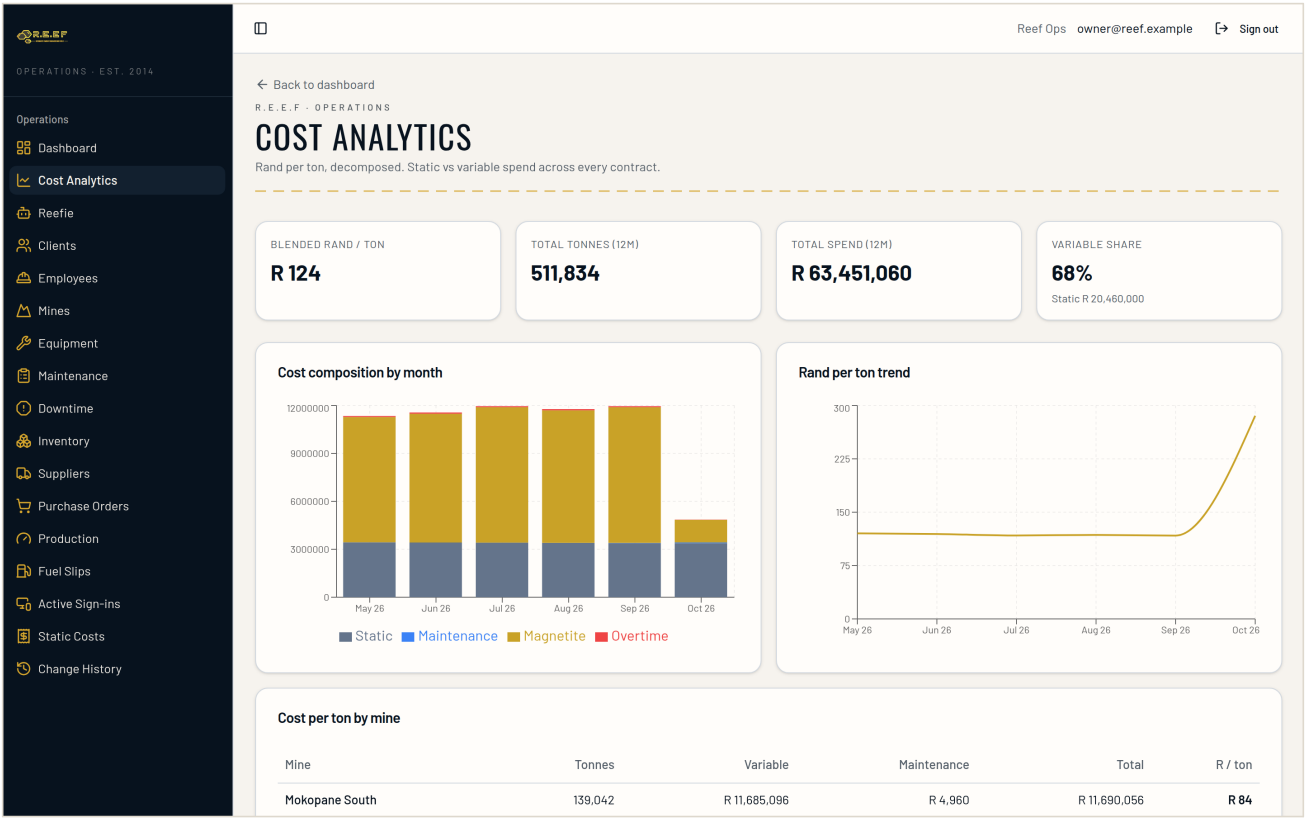



*Figure 27: Cost Analytics. The bar chart is the one that answers why, rather than only what.*

| COST | WHAT IS IN IT | FIXED OR VARIABLE |
|---|---|---|
| **Static costs** | Salaries, leases, insurance, licences | Fixed |
| **Magnetite** | The dense mineral the wash plant separates coal with | Variable |
| **Overtime** | Hours worked beyond the shift | Variable |
| **Maintenance** | Labour and parts on every repair | Variable |
| **Fuel** | Diesel drawn against a machine at a site | Variable |

NOTE

A period with no production has no cost per ton at all. It is left empty rather than shown as zero. A site that produced nothing still cost money, and a zero would read as free and sort to the top of a cheapest first list, which is the opposite of the truth.

## 12.2 The month end report

One month, one site, in the shape REEF reads it: cost per ton with its parts, production day by day, downtime grouped by cause, and a short written summary. It can be read on screen and downloaded as a spreadsheet for sending on.

| PART OF THE REPORT | WHAT IT CONTAINS |
|---|---|
| **Cost per ton** | The figure, with fixed and variable costs separated. Fixed costs are apportioned by days, so a part month is charged a part month |
| **Production** | Tons for each day of the month |
| **Downtime** | Hours by cause, largest first |
| **Narrative** | Two or three sentences built from the figures above, naming the largest cause of downtime |

NOTE

The narrative contains no figure that is not in the report beneath it. It is built from those figures rather than written separately, so the words and the numbers cannot drift apart.

WARNING

A report can go out of date, and the system says so. Producing a report is noted. If an entry dated inside that month is added, changed or removed afterwards, the report is marked out of date, because the figures in the copy somebody has already read are no longer the figures in the system. A month still running is never marked: it is expected to change.

## 12.3 The spreadsheet

The same report, downloadable as a file that opens in Excel. One row per day, with the month's cost figures repeated on every row so that a row still makes sense once it has been sorted or filtered out of its order.

TIP

Two things in that file exist because of Excel rather than because of REEF. It starts with a marker that tells Excel the file is Unicode, without which a name with an accent in it arrives mangled. And a cell that begins with an equals sign is written as text, so a note a worker typed cannot become a formula Excel decides to run.

# 13. Reefie, the assistant

Asking in plain words about stock, repairs, tons or costs, instead of opening four screens and doing the arithmetic.

1. From the menu, select Reefie.
2. Select New chat, or carry on with a conversation down the left.
3. Type the question. How much magnetite is left at Mokopane, and what did we spend on repairs last month, are both the kind of question it answers.
4. Read the answer. While it works it says it is checking the site data, because it is reading the records rather than recalling anything.

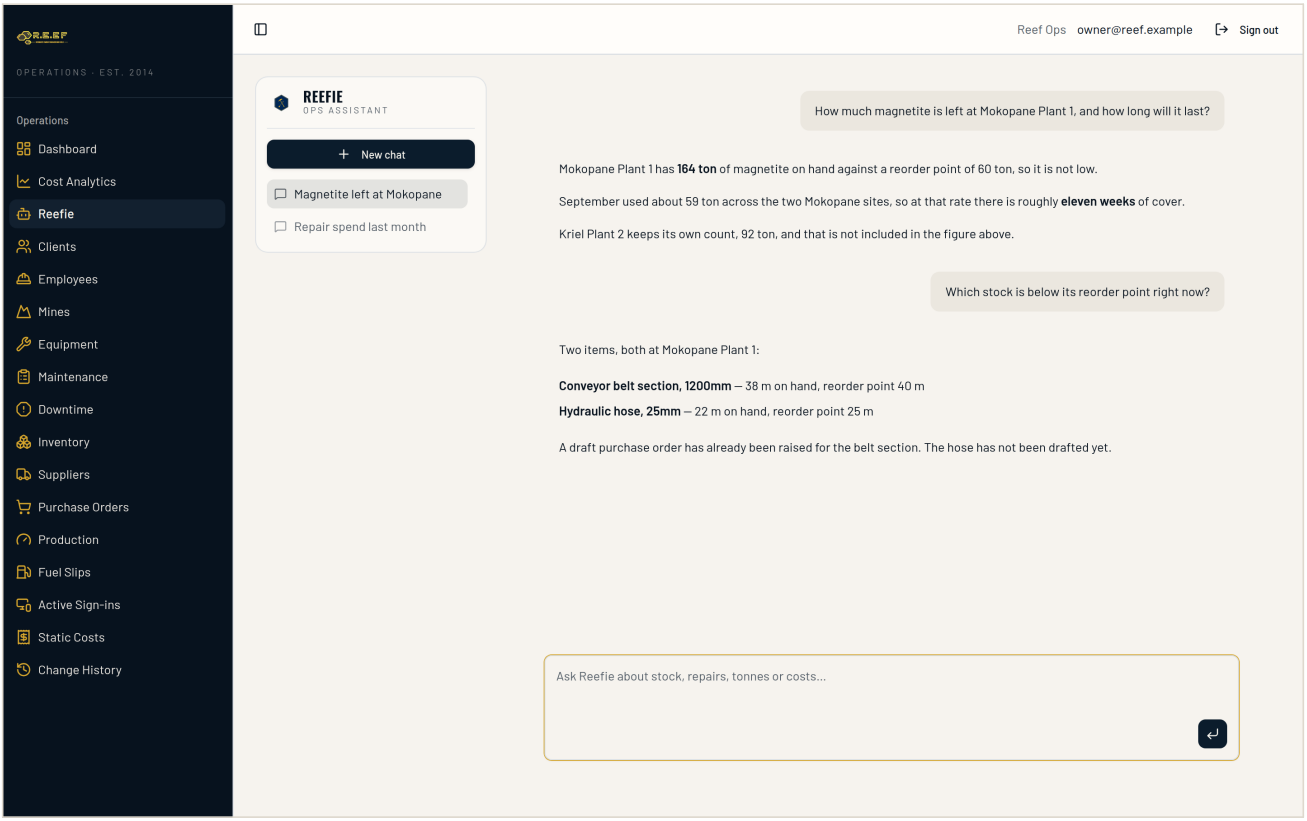



*Figure 28: A Reefie conversation. Earlier conversations are kept down the left.*

## 13.1 Why the figures can be trusted

The assistant drafts a sentence from figures it has just read out of the system. Before that sentence reaches you, every number in it is checked back against the figures it was built from. If a single number cannot be matched, the whole sentence is thrown away and you are given the raw figures instead, with a note saying what happened.

That is the difference between asking a model to be careful and making it impossible for an invented figure to reach a reader unremarked. Every check is recorded, so how often it happens is a figure REEF can see rather than a claim the team makes.

NOTE

Reefie will not compare one site against another. It refuses before it reads any figures. REEF's reason is that each plant has its own management team and one team's figures are not the other team's business. For a site by site ranking, the owner uses the comparison in Cost Analytics.

TIP

Ask one thing at a time, and name the site and the month. Magnetite at Mokopane in September gets a precise answer; how are we doing gets a vague one.

# 14. Service reminders and notifications

Every other figure in the system exists because a person captured it. A machine falling due for a service is a date passing, and nothing is written when a date passes. The reminder is therefore the one thing that happens with nobody asking for it.

## 14.1 What makes a machine due

| | THE RULE |
|---|---|
| **By date** | The next service date on that machine's most recent repair has been reached. With no repair recorded, the service interval counted from the last service, or failing that from the install date |
| **By tons** | The tons the machine has carried since it was installed have reached the next service tonnage, or its tonnage interval |
| **Which wins** | The date is checked first, so a machine late on both reads as late rather than busy |
| **Never due** | A machine that is not active, and a machine with no interval and no next service recorded |

> **NOTE**
> That last row is deliberate. A reminder raised against a figure nobody entered teaches people to ignore reminders, and an ignored reminder is worse than none. If a machine should be reminding you and is not, give it a service interval on the Equipment screen.

## 14.2 Who is told

Owners and managers. A worker cannot book a machine in, so a reminder they can do nothing about is noise. Managers are told only about machines at their own plant. A machine whose plant nobody has recorded reaches the owner alone, deliberately: they are the person who can fix the record.

The sweep runs once a day on South African time and claims the day before it starts, so running it twice raises nothing the second time. The same reminder is never sent twice.

## 14.3 Reading your notifications

1. Open your notifications from the top of the screen.
2. A service reminder names the machine and the date it falls due.
3. Mark one read once you have acted on it. Reading it is not acting on it.

> **WARNING**
> A sweep that does not run raises no error anywhere. Reminders simply stop arriving, which looks exactly like a month with no services due. The owner can see the days it ran and the days it did not; that list is in the administrator guide, and it is worth a look after any deployment.

# 15. Your account and your devices

Everybody can see which devices are signed in to their own account, and sign one out without changing their password. A handset with the application installed appears in this list the same way a browser does.

1. A manager or the owner selects the active sign ins item at the bottom of the menu. A worker selects the same tile on the home screen. It is spelt with a hyphen on the screen itself.
2. Read the list: the browser and device, the address it connected from, and when it was last used.
3. Select Sign out this device on anything you do not recognise.

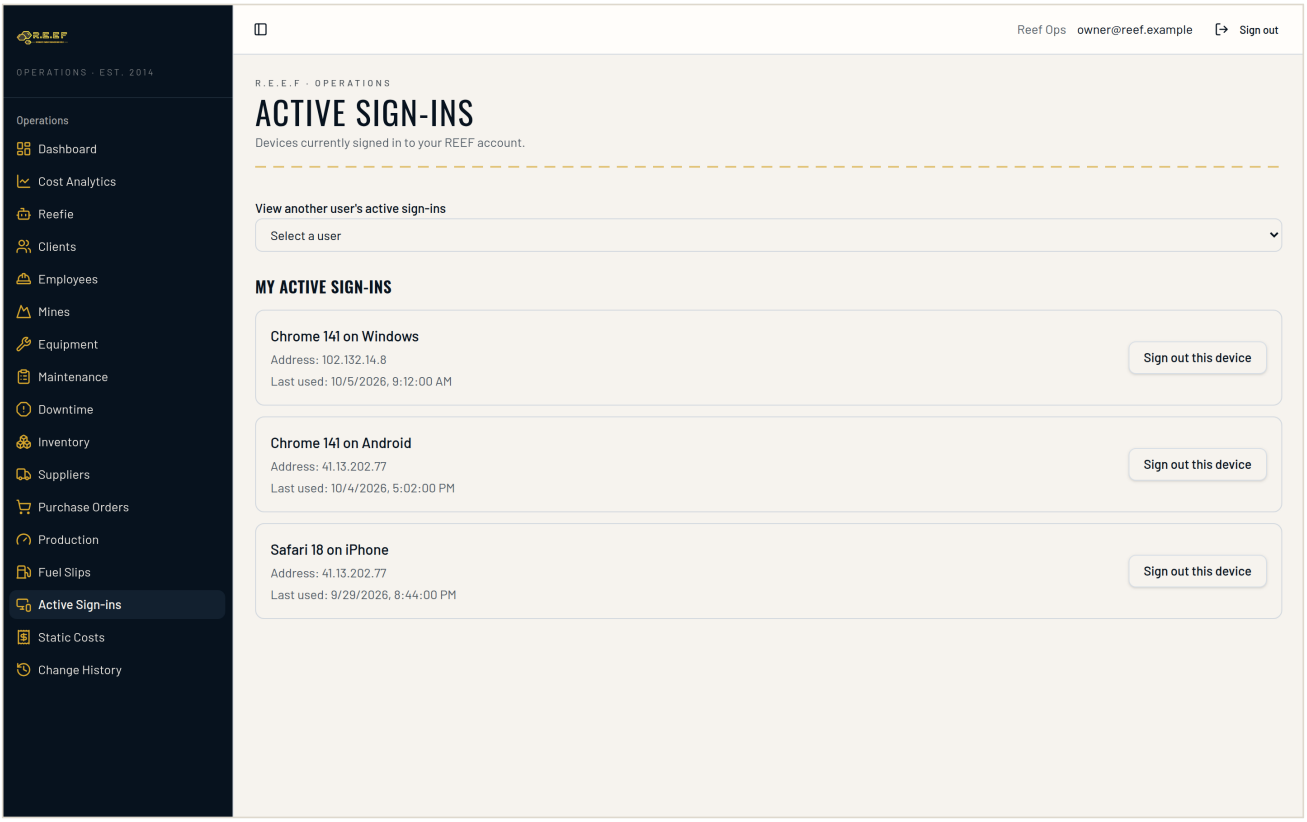



*Figure 29: Active sign ins. The owner also has a list at the top for looking at somebody else's devices.*

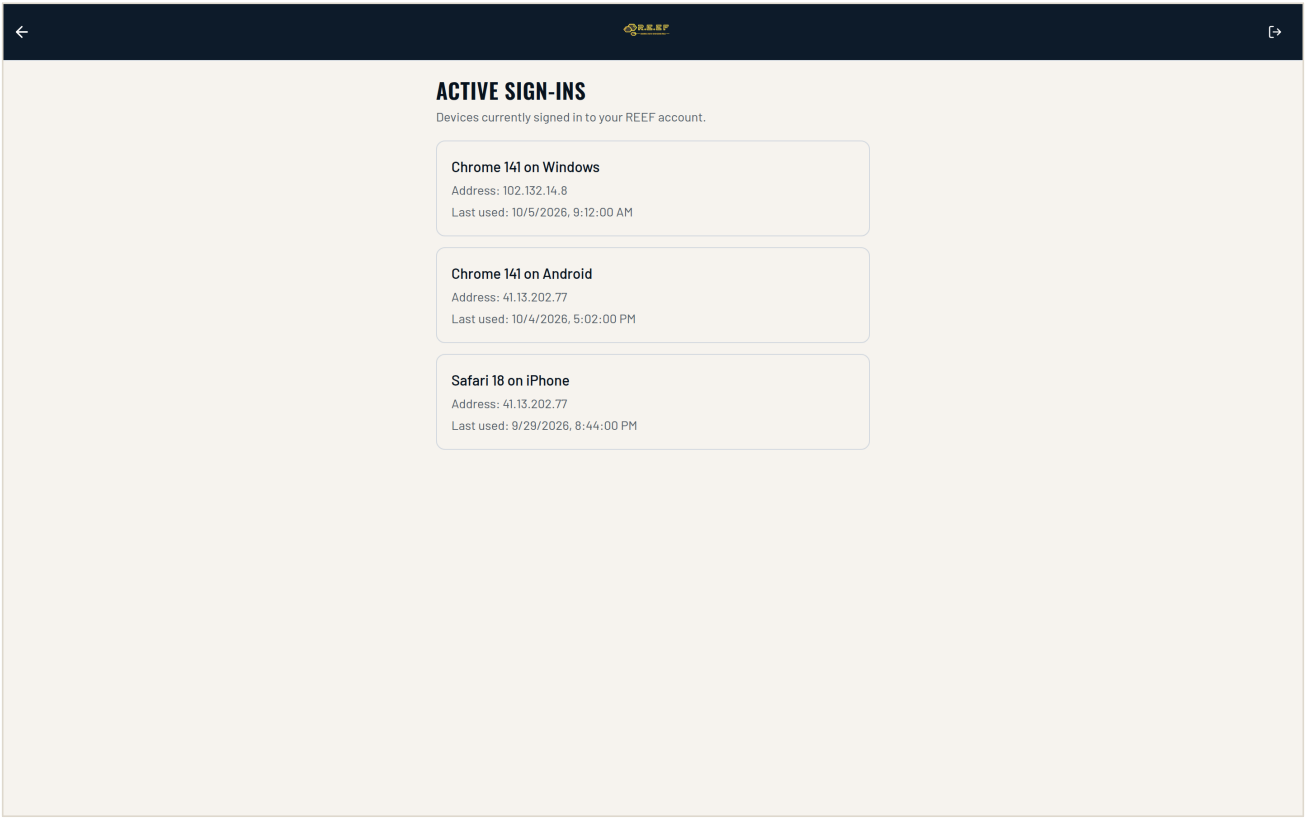



*Figure 30: The same list on the worker screens.*

**WARNING**

If a phone or laptop is lost, say so at once. Signing the device out stops it being used, immediately and everywhere, without anybody having to change a password. Tell your site manager, who tells the owner. The owner can sign out one device or every device on an account.

**NOTE**

Each sign in is checked on every request, not only when you sign in. A device that has been signed out stops working on its very next action, rather than when its session happens to expire.

# 16. When something goes wrong

Only problems this system actually produces. If what you are seeing is not here, section 19 says who to ask and what to tell them.

| WHAT YOU SEE | WHAT IT PROBABLY IS | WHAT TO DO |
|---|---|---|
| **The page will not load, or a wheel spins and nothing happens** | No signal, or signal too weak to reach the server | Check whether another website loads. If nothing loads, move to where there is coverage. If other sites load and this one does not, tell your site manager |
| **You're offline. REEF needs a connection to load** | The handset application cannot reach the system | Move to where there is coverage and select Try again. Nothing you had typed is kept, so write the figures down first if you were part way through |
| **Cannot reach the REEF server. Check your connection and try again** | The browser could not reach the system at all | The same check. If you are in coverage and it persists, it is the system and not you |
| **Your session has expired or is not valid. Sign in again** | You were signed in a long time without using it | Sign in again. Anything you saved is saved. Anything still on the screen unsaved is not |
| **This sign in has been revoked. Sign in again** | That device was signed out, usually by the owner | Sign in again if it was expected. If it was not, say so |
| **Your role does not allow this** | You are doing something your role does not cover | Check section 4.2. If you should be able to, your role may be wrong, and only the owner can change it |
| **A reason is required when changing a record** | The reason box was left empty | Say why in one line and save again. The box is on every edit dialog |
| **Someone else changed this after you opened it** | Two people had the same record open | Read their version, which the dialog shows you, then load it and save again. Section 11.2 |
| **This entry is dated ... which is ... days ago** | The entry is older than the capture limit, sixty days unless the owner changed it | Check the date for a mistyped year first. If the date is right, the entry is too old to capture and your site manager has to decide what to do about it |
| **This entry is dated ... which is after today** | A date in the future, almost always a mistyped year | Check the year. An entry cannot be dated ahead of today, because it would sit in a month that has not happened yet and pull that month's cost per ton out of shape |
| **Your stock list is empty, or shows parts you have never seen** | The plant on your account is wrong, or has not been set | Tell the owner. You cannot change your own plant and you should not use somebody else's account |
| **That stock item does not exist, or it isn't at your plant** | A link to an item at another plant | Nothing is broken. The system answers the same way for an item that does not exist and one you may not see, on purpose |
| **The stock figure does not match the shelf** | Something was taken without being recorded, or recorded twice | Tell your site manager. Do not adjust the figure to match the shelf: the difference is the useful information and correcting it quietly throws it away |
| **This account is not linked to an employee, so a purchase cannot be recorded against it** | Your login has not been joined to your employee record | Ask the owner to link them. A claim is paid to a person rather than to a login, so the system will not accept one it cannot attach to somebody. It is a one off fix |
| **A claim you filed is not in the Reimbursements list** | You are looking at the wrong tab, or the claim was paid or voided | The screen opens on pending. Select the all tab, or clear the month box. A voided claim carries the reason it was voided when you open it |
| **Reefie answered with raw figures instead of a sentence** | The number check found a figure in the draft that did not match the records | Use the figures, which are correct. Mention that it happened |

| WHAT YOU SEE | WHAT IT PROBABLY IS | WHAT TO DO |
|---|---|---|
| **Reminders have stopped arriving** | The daily sweep has not been running | Ask the owner to check the job list. A sweep that does not run reports nothing anywhere else |

# 17. Questions people ask

| QUESTION | ANSWER |
|---|---|
| **Is there an app I can install?** | Yes, on Android. Your site manager has the installation file and section 7 walks through it. It is not in the Play Store in this release. On an iPhone, use Safari and add the address to your home screen, which works and behaves much the same |
| **Can I capture with no signal?** | No, not yet, in the browser or in the application. The application tells you plainly that you are offline rather than losing what you typed. Write the figures down and enter them when you are back in coverage, the same day. Holding a shift on the handset is designed and is the next thing being built |
| **Is the application a different system?** | No. It opens the same system you reach in a browser, so it is always the same version as the office and there is nothing new to learn |
| **I paid for something myself. How do I get the money back?** | Select Money I Spent on the worker home screen, say what you bought, pick a category, enter the amount and photograph the slip. It waits as pending and the owner pays it at month end. Section 6 walks through it |
| **Do I have to attach the receipt?** | No, but attach it if you kept it. A claim with a photograph is usually read once and paid. A claim without one gets queried first, which is how a claim ends up paid a month later |
| **Who decides whether my claim gets paid?** | The owner. A site manager can see the claims at their own plant but cannot pay or refuse one. If a claim is refused, the reason is recorded and you can be told what it was |
| **Why does the system not ask me for the date?** | Because an entry is meant to be captured on the day. It is dated today in South African time. If you have to record something from yesterday, a manager can correct the date |
| **Can I change something I captured yesterday?** | Ask your site manager. A correction carries a reason that is kept, and that is a manager's screen |
| **Why do I have to give a reason when I correct something?** | So that a year later a corrected figure and a figure somebody quietly changed do not look the same |
| **Can I book a part out if I am not a manager?** | Yes. REEF asked for that. You may also see what is on the shelf. What you may not do is place the order for a replacement |
| **Does booking a part out order another one?** | No. It raises a reorder request when the figure runs low, and a manager turns that into an order. A request is not an order and nothing has been bought |
| **Why can I not see the other plant's stock?** | REEF asked for that too. Each plant is managed by its own team and keeps its own count. The same part at two plants is two figures, not one shared one |
| **Why can my manager not compare our site with the other one?** | Comparing sites is the owner's. A manager sees every figure for their own site, including its cost per ton |
| **Why is rand per ton so high on the second of the month?** | The month's salaries and leases are in it and only two days of tonnage are. It settles as the month fills. The month end report does not have this effect |
| **Who do I ask for a password reset?** | The owner. Nobody can read your existing password, so it is replaced rather than looked up |
| **I have lost my phone. What happens to my account?** | Tell your site manager at once. The owner signs that device out, which stops it working on its next action, whether it was the browser or the application. Your password does not have to change |

# 18. Glossary

Words this manual uses that do not mean the same thing everywhere.

| TERM | WHAT IT MEANS HERE |
|---|---|
| **APK** | The installation file for an Android application. REEF's is given to you by your site manager rather than downloaded from a store |
| **Booking out** | Recording that a part, consumable or tool was taken off the shelf. It lowers the stock figure. It does not order a replacement |
| **Capture** | Recording something on a screen at the time it happened, rather than writing it on a slip for somebody else to type in later |
| **Capture limit** | How old an entry may be when it arrives. Sixty days unless the owner changes it. Separately, no entry may be dated after today |
| **Cost per ton** | Everything a site cost over a period, divided by the tons it produced in that period. The figure REEF runs the business on. Also called rand per ton |
| **Downtime** | Time production was stopped, recorded with how long and why |
| **Draft order** | A purchase order nobody has approved. Nothing has been bought |
| **Fixed costs** | Costs that do not move with how much is produced: salaries, leases, insurance, licences. Called static costs on the screens |
| **Handset application** | The REEF application installed on an Android phone. The same system as the browser, with its own icon |
| **Magnetite** | The dense mineral used to separate coal by density in the wash plant. A variable cost, because more tonnage uses more of it |
| **MTD** | Month to date. From the first of this month until today |
| **Plant** | One place REEF operates, and the thing that decides whose stock and whose orders you can see. A site is the mine itself; a plant can run more than one |
| **POPIA** | The Protection of Personal Information Act, the South African law governing how personal information is held and for how long |
| **Purchase order** | A commitment to buy from a supplier. Management only |
| **Queue** | The store on the handset that will hold entries captured with no signal. Designed and not yet built. Section 7.5 |
| **Reimbursement, or claim** | Money a worker paid out of their own pocket for the site, recorded so REEF can pay it back. It is pending until the owner marks it paid or voids it. Section 6 |
| **Reorder point** | The figure at which the system raises a reorder request. Set per item, per plant |
| **Reorder request** | What booking stock out raises when the figure runs low. A request is not an order. A manager turns one into the other |
| **Retention period** | How long information is kept before it is removed. There are two: five years for an identity number and seven for the rest of an employee record |
| **Role** | Worker, site manager or owner. It decides what kind of thing the system allows you to do |
| **Session, or sign in** | One device signed in to your account. Each one can be signed out on its own |
| **Sweep** | The job that runs once a day, finds the machines falling due for a service and raises the reminders |

| TERM | WHAT IT MEANS HERE |
|---|---|
| **Variable costs** | Costs that move with how much is produced: magnetite, overtime, maintenance and fuel |
| **Version** | A counter on every record that goes up each time it is changed. A save carrying an old version is refused, which is what stops two people overwriting each other |
| **Voided** | A reimbursement claim REEF is not paying, with the reason recorded against it. The claim is kept rather than deleted |

# 19. Who to ask

Ask in this order. Most questions are answered at the first step.

| IF YOU NEED | ASK | HOW TO REACH THEM |
|---|---|---|
| **Help using a screen, or a password reset** | Your site manager | In person, or on the site group |
| **The installation file for the handset application** | Your site manager | In person. Do not pass it on outside REEF |
| **A reimbursement claim that has not been paid, or was refused** | Your site manager, then the owner | At month end, with the claim open on the screen |
| **A new account, a role changed, a plant corrected, or an identity number** | The owner | [ name and contact ] |
| **A device signed out because something was lost** | Your site manager, then the owner | At once, by phone |
| **Something that looks like a fault in the system** | The project team, through the site manager | [ email address ] |
| **Anything about how long information is kept** | The owner, who holds REEF's instruction in writing | [ name and contact ] |

> **TIP**
> When you report a fault, four things make it fixable: what you were trying to do, what the screen said, the date and time, and whether it has happened before. A report saying it is not working takes three messages to turn into any of that.